

PROTON INTELISENS DGK-I4 SERIES INSTRUCTION MANUAL

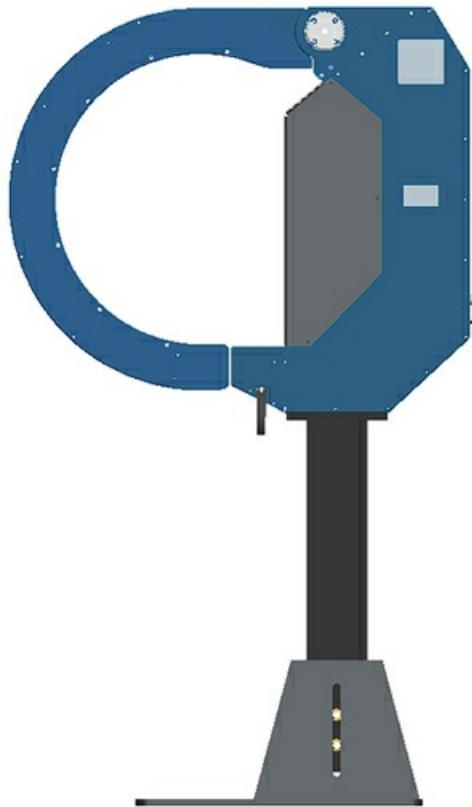


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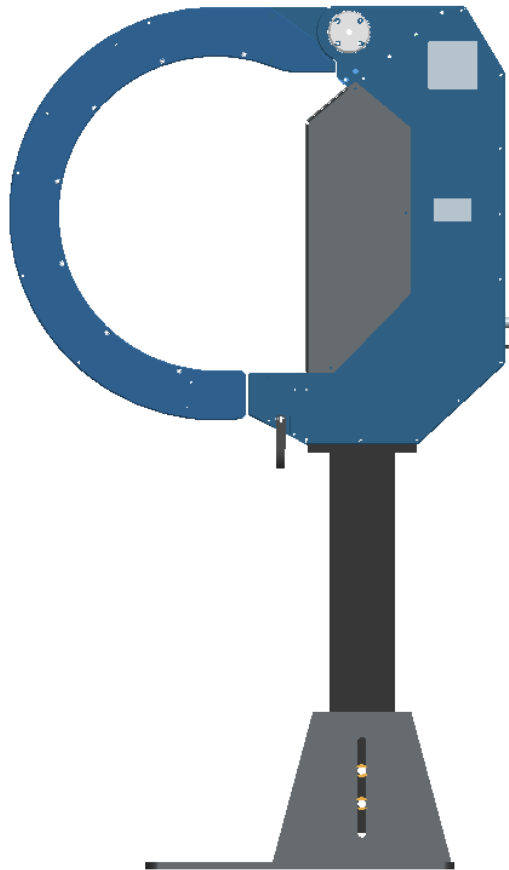
MANUAL FEEDBACK AND COPYRIGHT

Other ManualsLib Projects

Instruction Manual

NEW IntelliSENS DGk-i4 Series Diameter Gauge

Issue 1h
21 June 2022



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DECLARATION OF CONFORMITY (CE)

Proton Products International Limited declares that the equipment listed below fulfils the requirements of directive EMC: 2014/30/EU and Low Voltage: 2014/35/EU.

The following standards were applied:

EMC - EN 61326-1:2013

Low Voltage - EN 61010-1:2010 (Scientific and measuring instruments)

Equipment Covered

Product name	Description	Part number
DGk2200-i4	200mm Dual Axis CCD Diameter Gauge	00057MC013
DGk2120-i4	120mm Dual Axis CCD Diameter Gauge	00057MC011
DGk3120-i4	120mm Dual Axis CCD Diameter Gauge	00057MC012
DGk3200-i4	200mm Dual Axis CCD Diameter Gauge	00057MC014
SiDi-CDI4	Colour touch screen display interface	00049MC021

These products carry the CE Mark:



The manufacturer of the above named equipment is:

Proton Products International Limited
10 Aylesbury End
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ENGLAND

Proton Products is an ISO9001:2015 registered company.

The declaration is signed by:

Paul Sives

A handwritten signature in black ink, appearing to read 'P. Sives', written over a dotted line.



INTRODUCTION

The NEW IntelliSENS DGk-i4 series of dual and triple axis diameter gauges provide super- fast and highly accurate non-contact diameter measurement of cables or other objects manufactured in a continuous process.

Unlike traditional laser diameter gauges which use a mechanically scanned laser beam, the NEW IntelliSENS DG-k diameter gauge uses an entirely solid state technique. For each measurement axis, a LED is used to generate a broad, collimated beam of light which fully illuminates the measurement zone. The shadow generated by an object inserted into this light beam is imaged by a CCD and the resulting electrical signal digitally processed to yield a diameter reading. This solid state approach results in the following advantages over traditional laser diameter gauges:

- Much higher measurement speeds (not limited by mechanical scan time).
- Higher reliability and MTBF of the gauge as it suffers no wear from moving parts.
- No laser precautions required as a non-laser light source is used.

When fitted with the CDI4 interface display unit, the IntelliSENS DG diameter gauge can operate as a standalone unit.

RS-232, RS-422, RS-485, Ethernet, industrial standard communication interfaces (PROFIBUS, EtherNet/IP or PROFINET) and WiFi communication interfaces are installed as standard for straightforward connection to computers or PLCs.

Analogue and pulsed line speed inputs are fitted as standard to provide line speed data to the gauge for optional PI feedback control and SMFD (Single Measurement Flaw Detection). Proton Products range of SL and SLR laser speed gauges offer excellent speed measurement performance and are directly compatible with the DG diameter gauge speed inputs.

User configurable digital inputs are provided as standard to reset gauge measurements and trigger printing. User configurable relay outputs are provided as standard to signal gauge status, lump and neck detection and measurements exceeding preset error limits.

Optional analogue outputs can be specified for installation at time of manufacture for output of diameter and error measurements as analogue voltages.

An optional auto-adaptive PI feedback controller can be specified for installation at time of manufacture to allow the gauge to control insulation extruders or capstan drives for diameter control.

SPECIFICATIONS

Model	DGk2200-i4	DGk2120-i4	DGk3200-i4	DGk3120-i4	Units
Number of axes	2	2	3	3	-
Scan rate	1500	1500	1000	1000	scan/s/axis
Cumulative scan rate	3000	3000	3000	3000	scan/s
Update time	0.6	0.6	1	1	ms
Object diameter	5~180 mm	5~120 mm	5~180	5~120	mm
Optical gate diameter	200	120	200	120	mm
Accuracy	$\pm (50\mu\text{m} + 0.05\% \text{ of object diameter}) \mu\text{m}$				
Output resolution	0.01 μm				

COMMON SPECIFICATIONS

Common Specifications

Specification		
Operating temperature	5 ~ 45°C	
Environmental protection	IP65	
AC Power supply voltage	100~240VAC	
AC Power supply frequency	50~60Hz	
AC Power consumption	100W	
Light source	LED	
Air wipe	Integrated air wipe system	
Measurement units (user configurable)	millimetres (line speed: millimetres / minute, length: metres)	
	inches (line speed: feet / minute, length: feet)	
Measurement modes	Solid	Solid object diameter
	Glass	Transparent object diameter
	Helix	Twisted / braided multi-core cable envelope diameter

STANDARD INTERFACES

2x digital inputs	User configurable function	Reset, Print Activation	
	Maximum input voltage	24 Vdc	
4x relay outputs	User configurable function	Gauge OK, Upper tolerances exceeded, Lower tolerances exceeded, Single Measurement Flaw Detection (SMFD)	
	Isolated contact rating	Maximum voltage	24 Vdc
		Maximum current	1 A
Line speed inputs	Required for Helix mode or optional PI feedback controller operation		
	Analogue input	0 - 10 Vdc, user scalable	
	Speed pulse input	250 kHz max frequency, 30 V or 50 V max pulses (on two distinct inputs), user scalable	
Communication interfaces	RS232*, RS422, RS485, CANbus**, Ethernet, WiFi		
	Choice of any one of: PROFIBUS, EtherNet/IP or PROFINET (selectable via CDI4 or PCIS)		

*An optional RS-232-to-USB converter cable is available for connection to USB equipped computers.

**CAN-bus protocol is proprietary and reserved for connection to other Proton Products equipment such as CDI4 interface display unit.

OPTIONAL INTERFACES

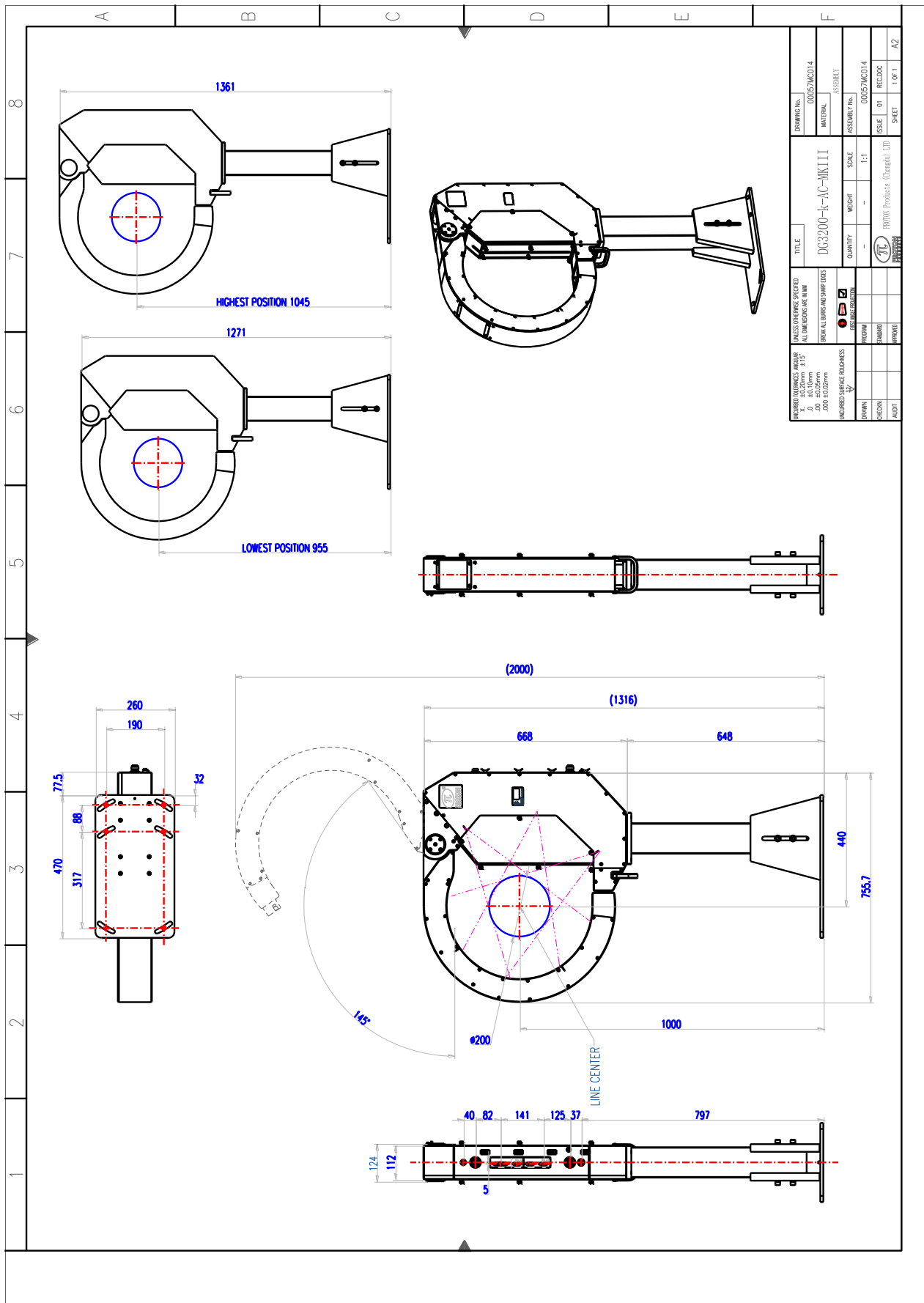
Must be specified for installation at time of manufacture, cannot be retrofitted.

3x Analogue outputs	±10 Vdc output of diameters or errors, user scalable
PI feedback controller	Proportional Integral feedback controller

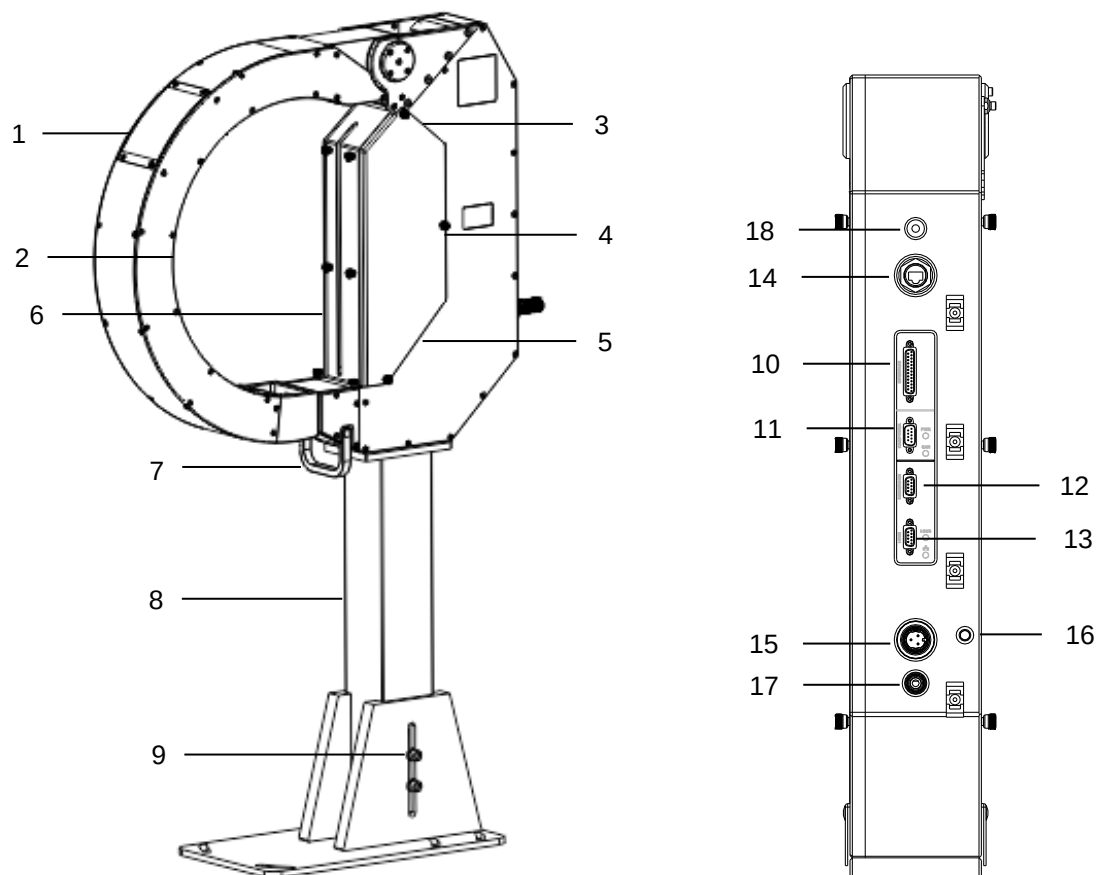
OPTIONAL FUNCTIONALITY

Statistics	Maximum, minimum, mean, standard deviation, C_p , C_{pk}
SPC	Statistical Process Control automatic set point for PI feedback controller (requires PI feedback controller option)
FFT	Fast Fourier Transform analysis for amplitude and frequency of periodic diameter variations
SMFD	Single Measurement Flaw Detection (Lump and Neck detection)

DIMENSIONAL DRAWINGS



ANNOTATED DRAWINGS

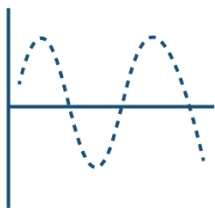
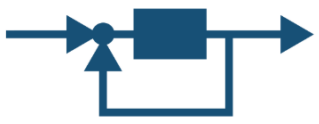


Label	Description
1	CDI4 interface display unit.
2	Optical window.
3	Y-axis light detector window.
4	Z-axis light detector window (DG3200/3120 only).
5	X-axis light detector window.
6	Detector window protection cover
7	Latch handle
8	Adjustable height stand
9	Height stand adjusting screws
10	DB25 interface
11	CAN/POWER
12	RS232/485
13	PROFI/EIP/DEV
14	RJ45 Ethernet socket
15	AC power inlet
16	Earth bolt
17	Air wipe air inlet
18	WiFi antenna

OPTIONAL ACCESSORIES

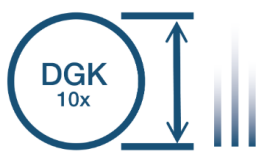
GAUGE ADD ON HARDWARE

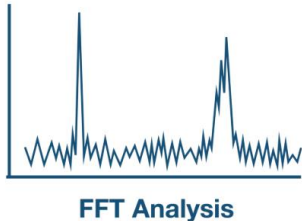
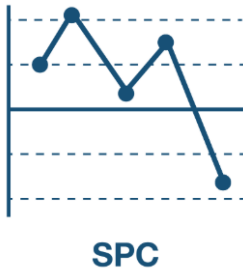
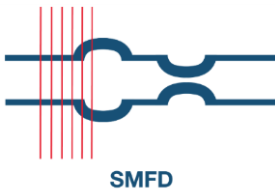
Must be specified for factory installation at time of ordering; cannot be field-retrofitted.

Analogue output	
 Analog Output	• 3x analogue ($\pm 10V$) outputs. • Independently programmable functions for X, Y, Z (DG3200/3120 only) diameters, average diameter, diameter errors from preset value, ovality and ovality error.
Proton part number	00054MC060
Proportional Integral (PI) feedback controller	
 Control Loop	• PI feedback diameter control via adjustment of insulation extruders or capstan drives. • Isolated $\pm 20V$ analogue input and $\pm 10V$ analogue output for connection to insulation extruders or capstan drives.
Proton part number	00054MC059

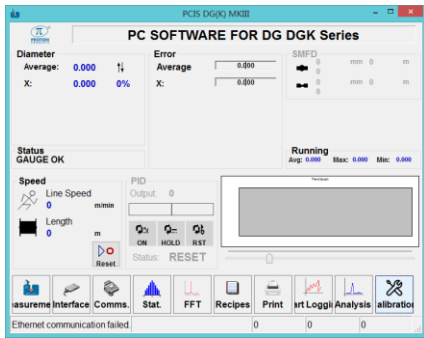
GAUGE ADD ON SOFTWARE

Optional software may be end-user enabled through the purchase of a license key. Some optional firmware functions are dependent on the presence of optional interfaces.

DGK 10K Scan Rate	
	• The DGK 10k Super Fast Scan option doubles the standard measurement rate to 10,000 measurements per axis per second. 30,000 measurements per second on a 3 axis gauge. Each scan takes just 1mS to obtain and output a new diameter reading. • DGK 10k Super Fast is ideal when combined with the SMFD or FFT Analysis options. Faster measurement rates identify smaller flaws and higher frequency periodicity.
Proton part number	00057SW010

Fast Fourier Transform (FFT) Analysis	
 Statistics	- Statistical data is collected by time, length, or reel. The sample period remaining is visible. The sample data is analyzed and displayed while the next sample data is collected. - Two selectable statistics channels are available. Max, Min, Mean and Standard Deviation along with Cp and Cpk Process Capability values are shown as values on the SiDi CDi4 Universal Display and graphically on the PCIS Software.
Proton part number	00057SW012
Fast Fourier Transform (FFT) Analysis	
 FFT Analysis	- Calculates FFT frequency spectra of measured diameters. - Identifies periodically occurring diameter fluctuations caused by production line problems. - Essential for telecommunications cable lines where periodic diameter fluctuations can degrade cable bandwidth.
Proton part number	00057SW013
Statistical Process Control (SPC) automatic set point control	
 SPC	- Minimises insulation material consumption whilst holding the insulation diameter in tolerance. Requires the PI feedback controller interface option to be installed.
Proton part number	00057SW011
Single Measurement Flaw Detection (SMFD)	
 SMFD	High-speed lump and neck flaw detection.
Proton part number	00057SW014

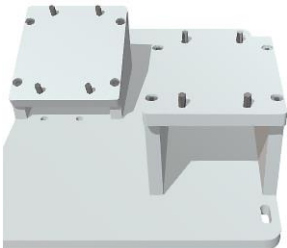
PC SOFTWARE

PCIS_DGK	PC Interface Software
	• PC-based software package. • User-friendly graphical user interface. • Displays all measurements. • Provides menu-based setting of all parameters. • Provides data logging, presets and alarms. • Connection via RS-232, Ethernet or EtherNet/IP.
Proton part number	00054SW015

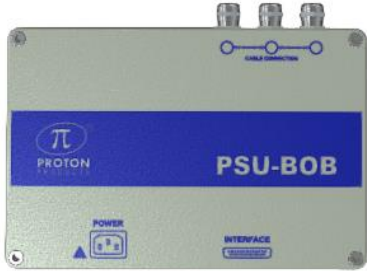
INTERFACE DISPLAY MODULES

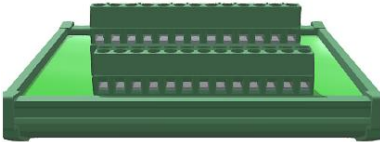
SiDi-CDi4	Touch screen display module		
	• Touch screen display. • Connects via the CAN-bus port. • Displays measured speed and length. • Provides menu-based setting of all parameters. • Mounts directly on the gauge body or remotely via an extension cable.		
Proton part number	00049MC023		
CAN-bus extension cable	Name	Length / m	Proton part number
	CAN 9DD_003M	3	00041CB003
	CAN 9DD_005M	5	00041CB005
	CAN 9DD_010M	10	00041CB010
	CAN 9DD_020M	20	00041CB020
	CAN 9DD_030M	30	00041CB030
	CAN 9DD_040M	40	00041CB040

HEIGHT STANDS

HST3	Heavy-duty adjustable height stand
	• For all DG gauges. • Heavy-duty steel tube and plate construction. • Must be bolted to the floor via four fixing holes. • Adjustable range for top of HST3: 700mm-1100mm
Proton part number	00009MC550
Adjustable Base DGK2200	Heavy-duty adjustable height stand
	• For DGK2200, DGK2120 gauges. • Auto pivoting height stand. • Heavy-duty steel tube and plate construction. • Must be bolted to the floor via four fixing holes.
Proton part number	00057MC040
6 Plane Bracket (DG3200-k)	Heavy-duty adjustable height stand
	• The 6-axis bracket used for mounting 2 x DG3200K gauges • Heavy-duty steel plate construction.
Proton part number	00057MC039

POWER SUPPLIES AND CONNECTIVITY

PSU-BOB-DG	Power supply and break-out box		
	• Connects via the DB25 “INTERFACE” connector. • Supplies 24VDC electrical power to the gauge. • Provides screw terminal access to all “INTERFACE” connector input and output pins. • End user cables are sealed with three cable glands. • Input voltage range: 90 – 260 VAC @ 45 – 65 Hz. • Select the required length of DB25 cable from below.		
Proton part number	00047MC660		

Male DB25 Terminal Strip			
	• Breaks out the DB25 “INTERFACE” connector to a DIN rail mountable set of screw terminals. • Select the required length of DB25 cable from below.		
Proton part number	00043MC045		

DB25 “INTERFACE” port to PSU-BOB-DG			
	Name	Length / m	Proton part number
	DB25_003m(9.8ft) cable	3	00041CT003
	DB25_005m(16ft) cable	5	00041CT005
	DB25_010m(32ft) cable	10	00041CT010
	DB25_020m(65ft) cable	20	00041CT020
	DB25_030m(98ft) cable	30	00041CT030

INSTALLATION

PRECAUTIONS

Operating temperature

Specification	Minimum	Typical	Maximum	Units
Operating temperature	+5		+45	°C

- Do not operate the gauge in temperatures outside of the specified range.
- Do not install the gauge near high temperature surfaces or objects which may cause it to overheat.
- Operation of the gauge outside the specified temperature range may result in degraded measurement accuracy, malfunction or damage to the gauge.

Protect from impact

- The gauge contains delicate optical and electronic assemblies and must never be dropped or struck by other objects.
- The gauge must be securely mounted by its base to prevent toppling.
- Measured objects threaded through the optical gate cavity must be secured against contact with the optical windows or catching upon the gauge body. For cable production line applications, Proton Products can supply optional adjustable cable guides for securing the passage of a cable through the optical gate cavity.

Do not open or disassemble

- The gauge contains no user serviceable components.
- Loosening the gauge screws or removing its cover will invalidate the product warranty.

Periodic maintenance

- The physical condition of the gauge, CDI4 interface display unit and connecting cables should be checked periodically; if any damage is suspected, then the unit should be taken out of service for inspection and repair or replacement of damaged parts.

Optical windows

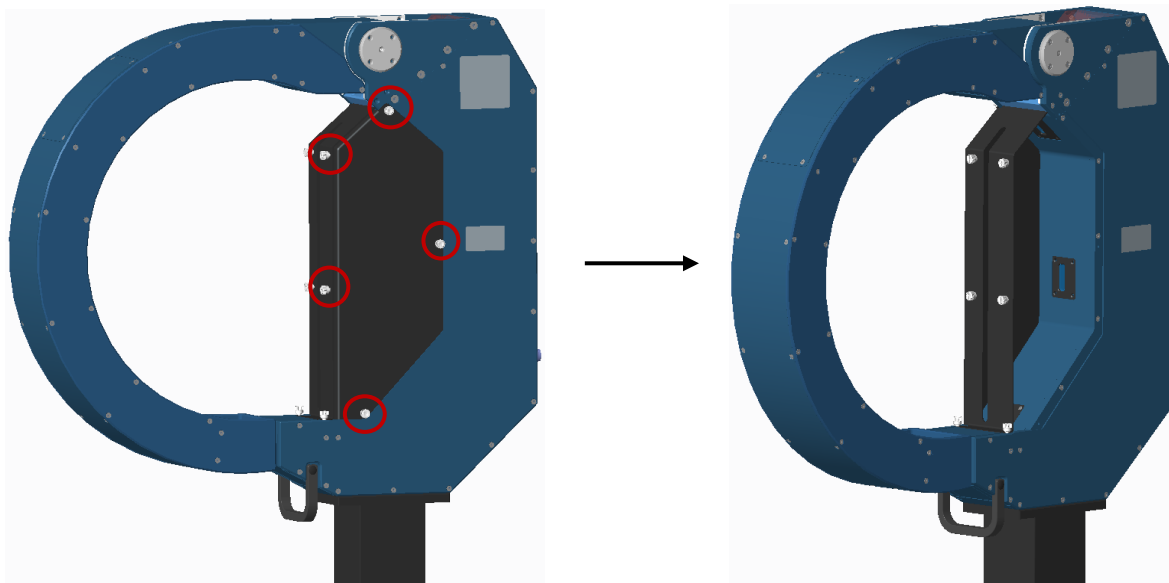
- Do not allow smoke, water, steam, dust or other debris to come into contact with any of the optical windows.
- Obstruction of the optical windows may degrade measurement accuracy or inhibit measurement.
- If any optical window appears to be damaged or misaligned, then the unit should be taken out of service for repair.
- If any optical window requires cleaning, then refer to the cleaning procedure detailed in this manual to minimise the risk of scratching the windows.

OPTICAL WINDOW CLEANING PROCEDURE

- The optical windows are manufactured from anti-reflection coated optical glass; they must be treated with the same level of care as a high-performance camera lens.
- **Before inspecting or cleaning the optical windows, ensure that the gauge is powered off and no light is emitted.**

Required items	Notes
Small blower brush	Such as the type used to remove dust from camera lenses.
Lens cleaning tissues or micro-fibre lens cleaning cloth	Do NOT use facial tissues as these can scratch delicate optics.
Lens cleaning solution	Such as the type specified for cleaning camera lenses.

1. Remove the lens protection cover by releasing the screws indicated below.



2. Use the small blower brush to remove any visible dust on the optical window.
3. Apply a few drops of lens cleaning solution to a fresh lens cleaning tissue or a clean micro-fibre lens cleaning cloth.
4. Gently wipe the optical window from the centre outwards; apply only light pressure to the tissue or cloth when wiping the optical window.
5. Repeat as necessary with fresh tissues or a clean section of cloth until the optical window is clean and free of all smears and smudges.

INSTALLATION SEQUENCE

Unpack the gauge and check for missing accessories and shipping damage.

Mechanical installation:

1. Mount the gauge securely on the height stand.
2. Install optional Proton Products cable guides.
3. Install air wipe air supply.

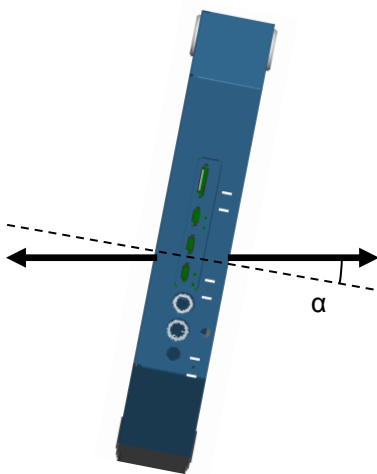
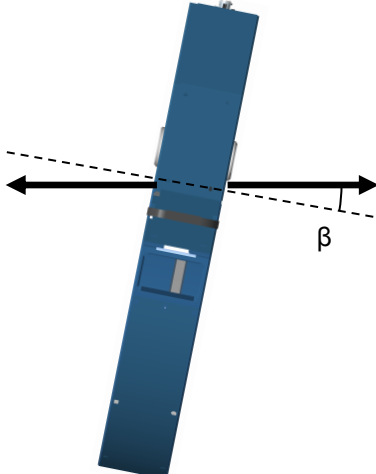
Electrical installation:

1. Install earth connections.
2. Install an AC power cable.
3. Install interface connections (RS-232, RS-422/485, Ethernet or optional PROFIBUS, EtherNet/IP, PROFINET or WiFi antenna).
4. Install speed input (analogue or pulse).
5. Install optional electrical interfaces (Logic inputs, Relay outputs, Analogue outputs, PI feedback controller connections) using the optional Proton Products PSU-BOB breakout box or terminal strip.
6. Configure the gauge either via the CDI4 interface display unit or any of the above interfaces.

MECHANICAL INSTALLATION

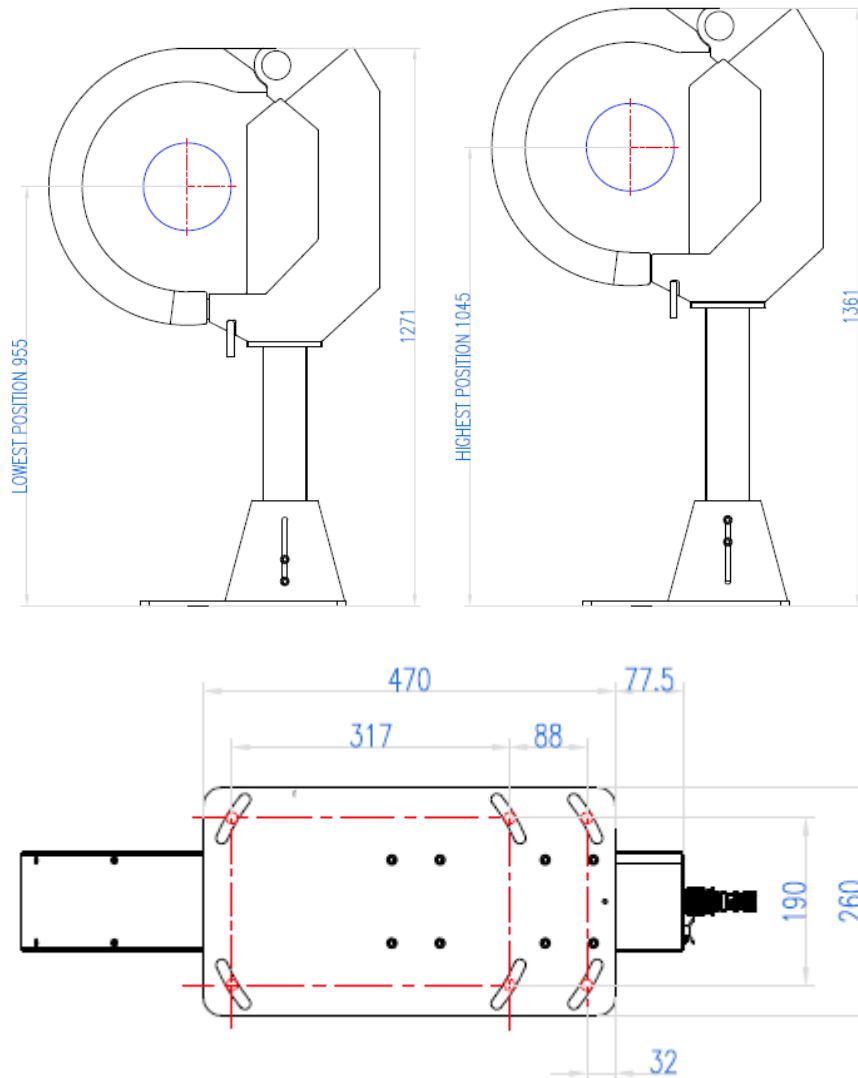
The gauge must be securely mounted by its base to a fixed surface. It can either be secured to an end user supplied mount or to an optional Proton Products adjustable height stand.

- Optimum measurement accuracy is achieved if the object is centred in the optical gate cavity.
- Misalignment of the gauge measurement axis relative to the object will result in the measured diameter being larger than the true diameter; for optimum accuracy, ensure that the pitch (α) and yaw (β) angles are as close to 0° as possible; this will correspond to the position for the minimum measured diameter:

Front view	Underside view	
		
Pitch (α) or yaw (β) angle misalignment cosine error		
$\text{Measured diameter} = \frac{\text{True object diameter}}{\cos(\alpha \text{ or } \beta)}$		
Pitch (α) or yaw (β) angle / °	($\cos \alpha$ or β)	Diameter error / %
0	1.000000	0
0.25	0.999990	+0.001
0.5	0.999962	+0.004
0.75	0.999914	+0.009
1	0.999848	+0.015
2	0.999391	+0.061
3	0.998630	+0.137
4	0.997564	+0.244
5	0.996195	+0.382
10	0.984808	+1.543

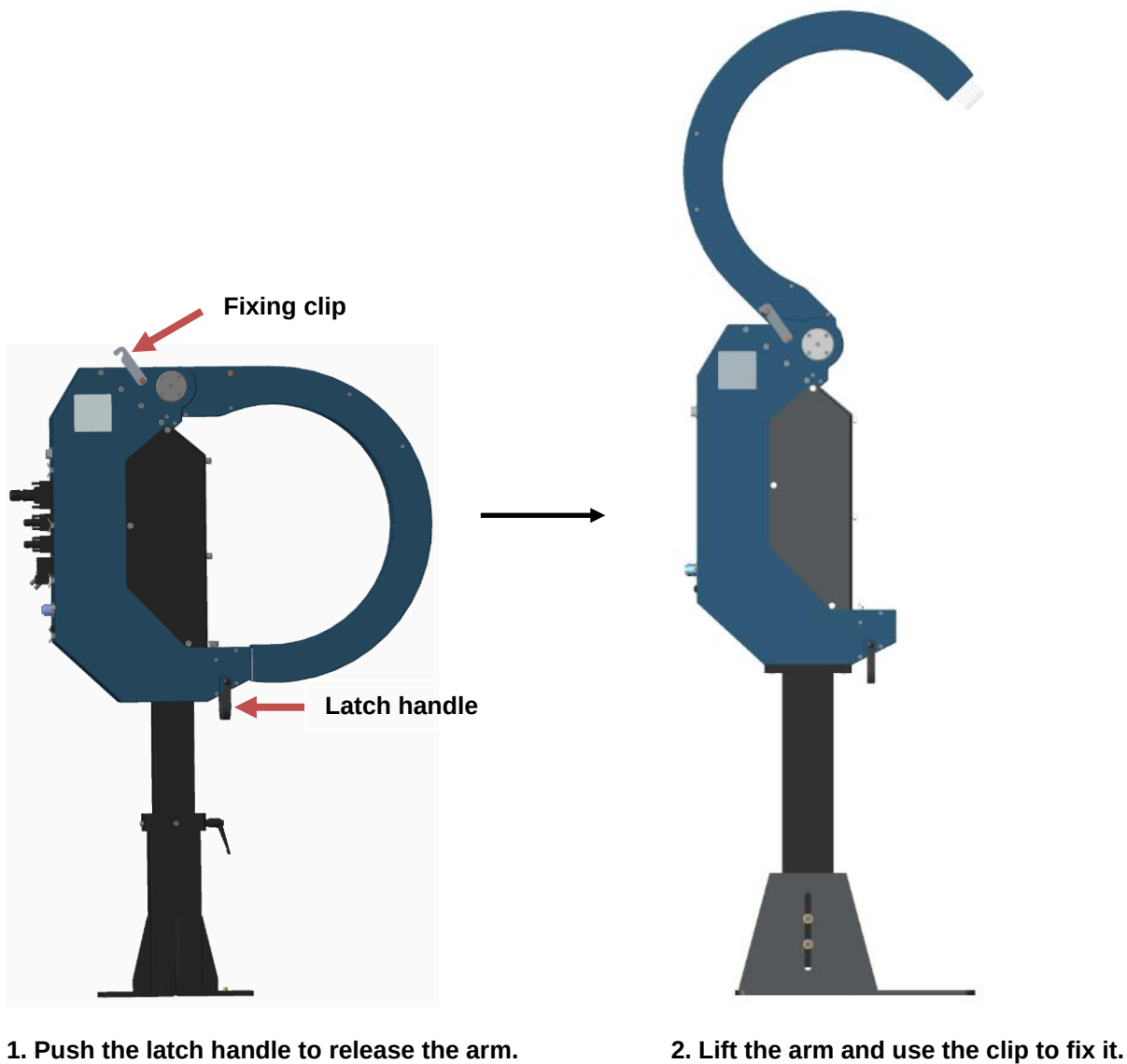
Installation on the adjustable height stand

Proton Products supplies an adjustable height stand which can position the measurement axis of the gauge within the height range from 955 to 1045mm. The height can be adjusted by releasing the adjusting screws.



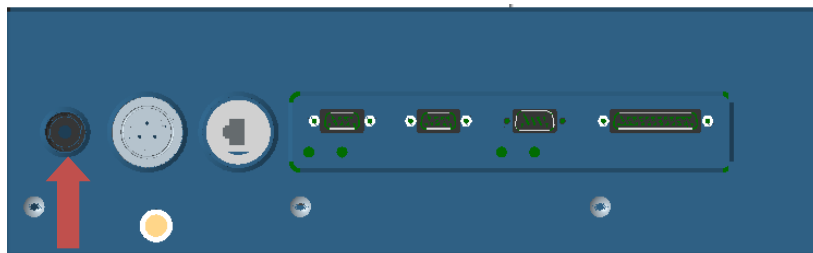
1. Mark and drill six holes into the floor for securing the base of height stand (appropriate floor mounting hardware is to be supplied by the end-user).
2. Securely bolt the height stand to the floor (the height stand is not designed for stable free standing operation).
3. Earth the height stand with an earth wire of at least 6mm² attached via a crimp on ring terminal to the M5 earth bolt on the base of the stand.
4. Fully extend height stand by loosening and tightening the adjusting screws as far as possible.
5. Attach the gauge to the stand using the four M8 x 20 mm screws supplied (do not exceed a torque of 6 Nm when tightening these screws).
6. Loosen the adjusting screws and adjust the height and alignment of the gauge to the object being measured.
7. Tighten the adjusting screws to lock the position of the height stand.

Arm Lifting Operation



Air wipe air supply installation

The air wipe system delivers a constant flow of clean air over the external surface of each optical window to keep it clear of dust, moisture, and debris.



Connector type: 6mm push fit

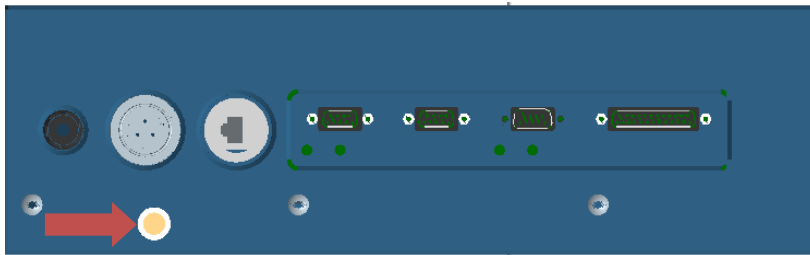
Specification	Minimum	Typical	Maximum	Units
Pressure		2	5	bar
Flow rate			140	litres / minute

Air quality must meet or exceed ISO 8573.1:2001 Class 1.3.1 (solids.water.oil):

Specification	Maximum	Units	Class
0.1 - 0.5 μm solid particle count	100	/ m^3	1
0.5 - 1 μm solid particle count	1	/ m^3	
1 - 5 μm solid particle count	0	/ m^3	
Water vapour pressure dew point	-20	$^{\circ}\text{C}$	3
Oil aerosol and vapour	0.01	mg / m^3	1

ELECTRICAL INSTALLATION

Earth connection



Connector type: M5 bolt

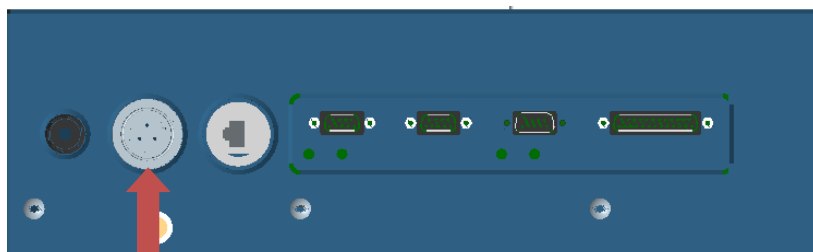
- An earth wire of at least 6mm² must be attached via a crimp on ring terminal to the dedicated M5 earth bolt on the case of the gauge.
- Do not rely on the mounting bolts to provide a reliable earth path.
- If a height stand is used then it must also be earthed via its own dedicated earth wire.
- All earth wires should be kept as short as practical.

Shielded Cables

- Use shielded cable for all signal connections.
- Ensure that all cable shields are correctly clamped and electrically connected to their connectors and metal connector shells at both ends.
- Ensure that the shields of cables connecting to the end user's equipment are clamped to earth at their destination.

POWER SUPPLY

Specification	Min	Typical	Max	Units
AC Power supply voltage	100		240	VAC
AC Power supply frequency	50		60	Hz
AC Power consumption (with CDI4 interface display unit)			100	W



Connector type: Bulgin Buccaneer 6000 Series PXM6010/03P/ST

Pin	Designation	Description	Notes
E	Earth	Earth	
L	Live	Live	
N	Neutral	Neutral	

Powering on the gauge

The gauge has no power switch; as soon as power is applied it will power up, perform self tests and then commence measurement.

Power indications

PWR		LED status	Indication
		Continuous red	Gauge is powered on
Power		Extinguished	Gauge is powered off

If an optional CDI4 interface display is attached to the gauge, then correct power up is indicated by the display of the welcome screen followed by the home page.

Powering off the gauge

The gauge has no power switch; it may be powered off by switching off or disconnecting the power supply to the unit.

CONFIGURATION VIA CDI4

Power on screen



When the gauge is powered on, the Proton welcome screen will be displayed momentarily.

Touch screen calibration

The CDI4 touch screen can be calibrated using the following procedure.

1. Press the “Click to calibrate touch screen” message during gauge power on to access the screen calibration function.
2. Tap the screen as directed, the screen will be calibrated.

Home page 1



- Gauge status / PID status (hold/ Run/ Reset) / PID output indication

- Average of measured X, Y, Z diameters

- Average error (average diameter – preset average diameter)

- Average error bar to indicate above value pictorially

Press the  button to reset the controller output voltage to zero.

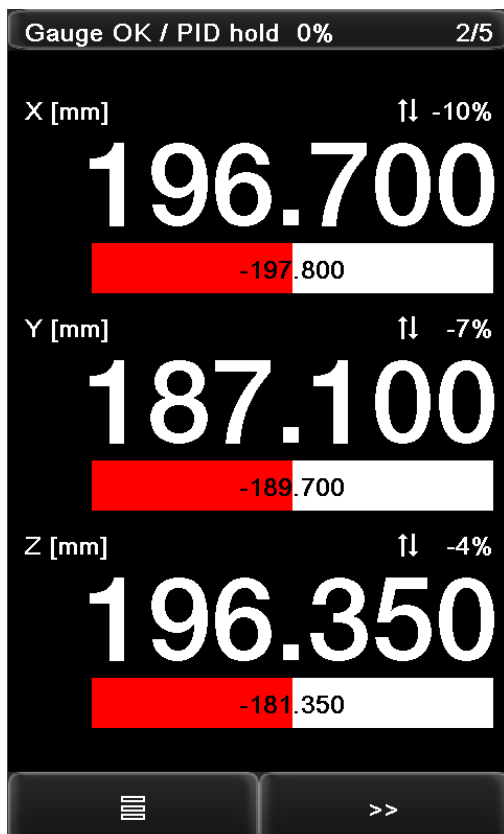
Press the  button to turn ON the controller output.

Press the  button to HOLD the controller output voltage constant at its current value.

Press the  button to access the function menu page.

Press the  button to advance to “Home Page 2”.

Home page 2



- Gauge status / PID status (hold/ Run/ Reset) / PID output indication

- Object position in X-axis optical gate (0% indicates object centred).

- Measured X diameter (time averaged).

- X error (X diameter – Preset X diameter)

- Object position in Y-axis optical gate (0% indicates object centred).

- Measured Y diameter (time averaged).

- Y error (Y diameter – Preset Y diameter)

- Object position in Z-axis optical gate (0% indicates object centred).

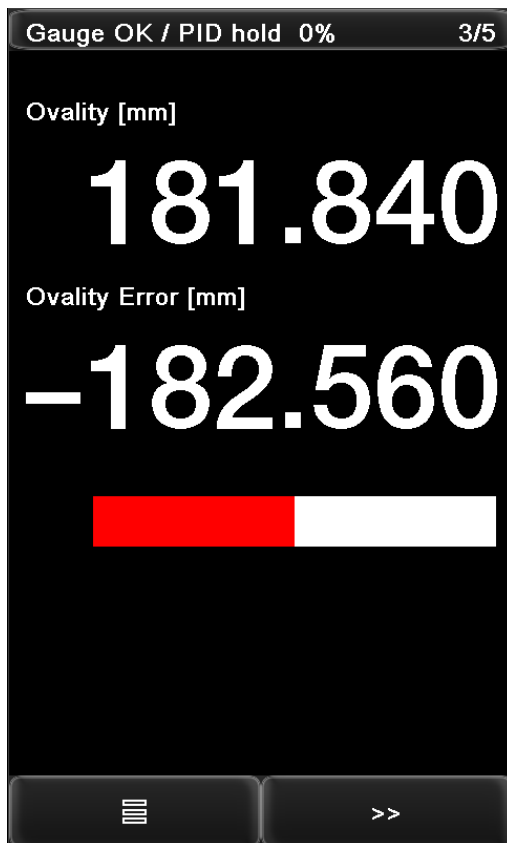
- Measured Z diameter (time averaged).

- Z error (Z diameter – Preset Z diameter)

Press the  button to access the function menu page.

Press the  button to advance to “Home Page 3”.

Home page 3



- Gauge status / PID status (hold/ Run/ Reset) / PID output indication

- Measured Ovality = $\max(X, Y, Z) - \min(X, Y, Z)$

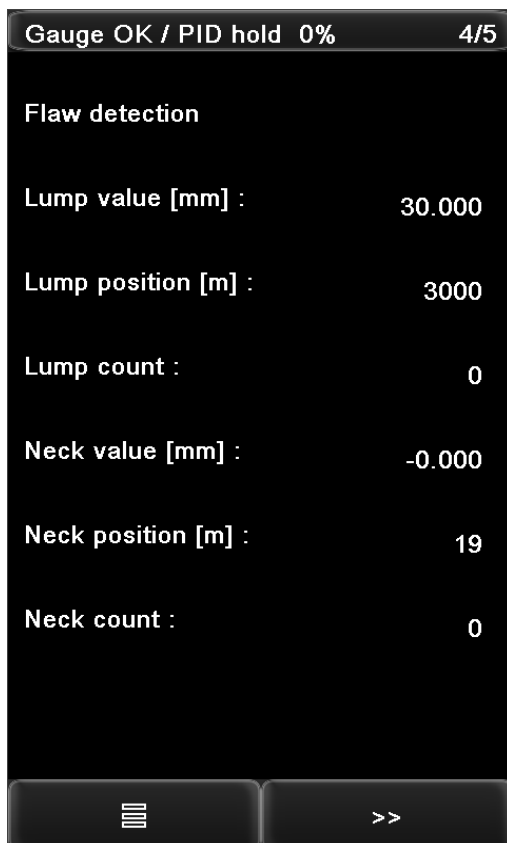
- Ovality error = Measured Ovality – Preset Ovality

- Ovality error bar to indicate above value pictorially

Press the  button to access the function menu page.

Press the  button to advance to “Home Page 4”.

Home page 4



- Gauge status / PID status (hold/ Run/ Reset) / PID output indication

- Size of the last lump flaw

- Position of last lump flaw

- Lump flaw count

- Size of the last neck flaw

- Position of last neck flaw

- Neck flaw count

Press the  button to access the function menu page.

Press the  button to advance to “Home Page 5”.

Home page 5



- Gauge status / PID status (hold/ Run/ Reset) / PID output indication

- Running average diameter

-Running maximum diameter

- Running minimum diameter

- Line speed (from one of the speed sources: user preset line speed; measured line speed from the speed pulse input; measured line speed from analogue input.)

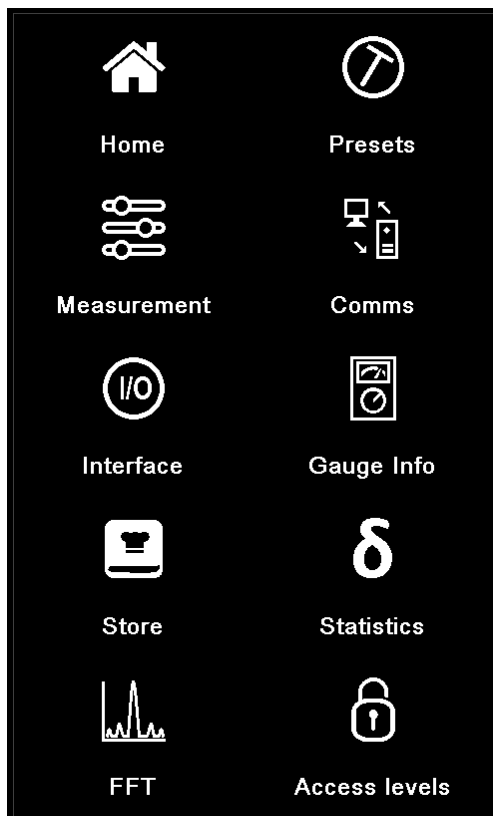
- Calculated cumulative length of object

Press the  button to reset above length to zero.

Press the  button to access the function menu page.

Press the  button to return to "Home Page 1".

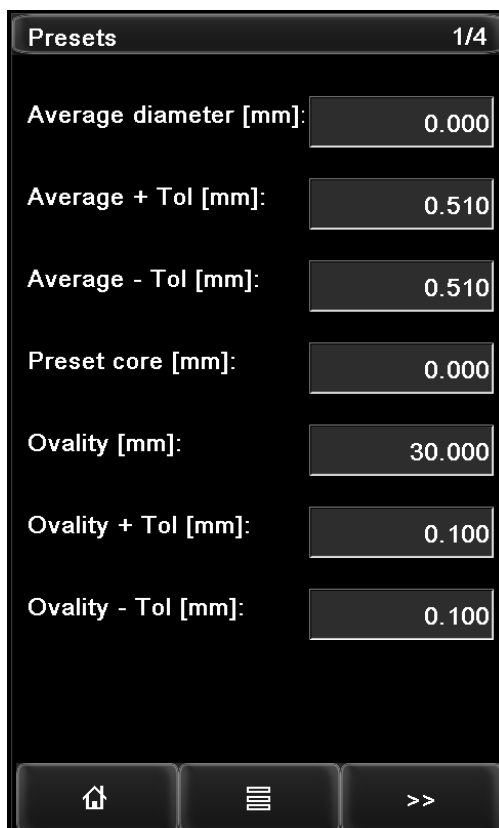
Function menus



Press the  button on the home page to access this page.

Press the **Home** icon to return to the home page; or press the other icons to access the respective configuration pages.

Preset



Press the **Presets** icon on the home page to access the Presets page 1.

- Set the preset average diameter.

- Set the preset average upper tolerance.

- Set the preset average lower tolerance.

- Set the preset core diameter (required for PI feedback controller operation).

- Set the preset ovality.

- Set the preset ovality upper tolerance.

- Set the preset ovality lower tolerance.

Press the  button to return to the home page.

Press the  button to access the function menu page.

Press the  button to advance to the next page.

Presets 2/4

X diameter [mm]:	25.000
X + Tol [mm]:	0.510
X - Tol [mm]:	0.510
Y diameter [mm]:	26.000
Y + Tol [mm]:	0.510
Y - Tol [mm]:	0.510





- Set the preset X diameter.

- Set the preset X upper tolerance.

- Set the preset X lower tolerance.

- Set the preset Y diameter.

- Set the preset Y upper tolerance.

- Set the preset Y lower tolerance.

Press the  button to return to the home page.

Press the  button to access the function menu page.

Press the  button to advance to the next page.

Presets 3/4

Z diameter [mm]:	26.000
Z + Tol [mm]:	0.510
Z - Tol [mm]:	0.510





- Set the preset Z diameter.

- Set the preset Z upper tolerance.

- Set the preset Z lower tolerance.

Press the  button to return to the home page.

Press the  button to access the function menu page.

Press the  button to advance to the next page.

Presets
4/4

Shrinkage mode:
Percentage

Shrinkage [%]:
0.0

SMFD tol mode:
ABS

SMFD + Tol [mm]:
0.000

SMFD - Tol [mm]:
0.000





- Select the shrinkage mode from options: Percentage; ABS.

- Set the percentage or absolute shrinkage value (depending on the setting for “Shrinkage mode” above). A value of zero disables shrinkage compensation.

- Select the SMFD tolerance mode from options: Percentage; ABS.

- Set the percentage or absolute SMFD upper tolerance value (depending on the setting for “SMFD tol mode” above).

- Set the percentage or absolute SMFD lower tolerance value (depending on the setting for “SMFD tol mode” above).

Press the  button to return to the home page.

Press the  button to access the function menu page.

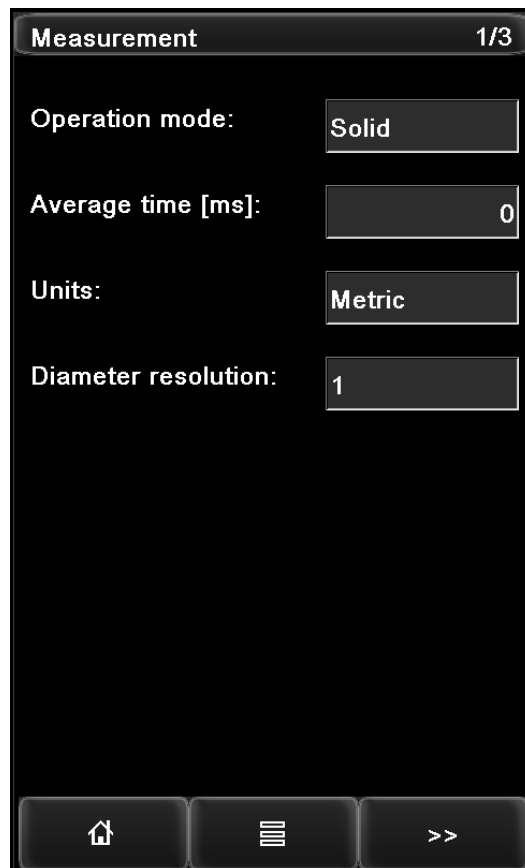
Press the  button to advance to the next page.

Shrinkage mode: Percentage; ABS

Percentage: percent shrinkage scales the averaged measured diameter by the Shrinkage.
Output diameter = (Measured diameter-Preset core) x (1– Shrinkage percentage)

ABS: absolute shrinkage mode deducts the Shrinkage from the averaged measured diameter.
Output diameter = (Measured diameter) – (Shrinkage)
(Note that if the calculated value is negative, the output diameter will be displayed 0.)

MEASUREMENT



Press the **Measurement** icon on the Menu page to access the measurement page 1.

- Select the operation mode from options: Solid; Glass; Helix.
- Set the time window over which diameter measurements are averaged.
- Select the measurement unit from options: Metric; Imperial.
- Select diameter resolution from options: 1; 0.1; 0.01; 0.001; 0.0001.

Press the  button to return to the home page.

Press the  button to access the function menu page.

Press the  button to advance to the next page.

Operation mode: Solid; Glass; Helix

Solid: Measure the diameter of opaque objects.

Glass: Measure the diameter of transparent objects such as glass rod (outermost edge detection).

Helix: Measure the envelope diameter that encloses a twisted or braided multi-core cable.

- Requires provision of line speed.
- SMFD (Single Scan Flaw Detection) is NOT available in “Helix” mode.

Units: Metric; Imperial

Metric: Millimetre (mm) and metre (m) measurement units.

Imperial: Inch (in) and feet (ft) measurement units.

Measurement / SMFD 2/3

Interval mode: Time

Interval [s]: 0.000

Measurement avg [ms]: 0.0

Reference avg [ms]: 0

Start speed [ft/min]: 0

Start length [ft]: 0





- Select the SMFD Interval mode from options: Time; Length.

- Set the SMFD Interval value (depending on the setting for "SMFD Interval mode" above).

- Set the SMFD measurement average time.

- Set the SMFD reference average time.

- Set the minimum line speed above which SMFD is enabled.

- Set a minimum length above which SMFD is enabled.

Press the  button to return to the home page.

Press the  button to access the function menu page.

Press the  button to advance to the next page.

Interval

After detection of a flaw, subsequent flaws detected within this time period (or within the length) will be ignored (subsequent flaws are effectively counted as part of the first flaw).

Measurement avg

Set the time window over which diameter measurements are averaged to generate the "Measurement Diameter".

Reference avg

Set the time window over which diameter measurements are averaged to generate the "Reference Diameter".

A flaw is detected if the above "Measurement Diameter" deviates from this "Reference Diameter" by more than the preset upper or lower SMFD flaw limits.

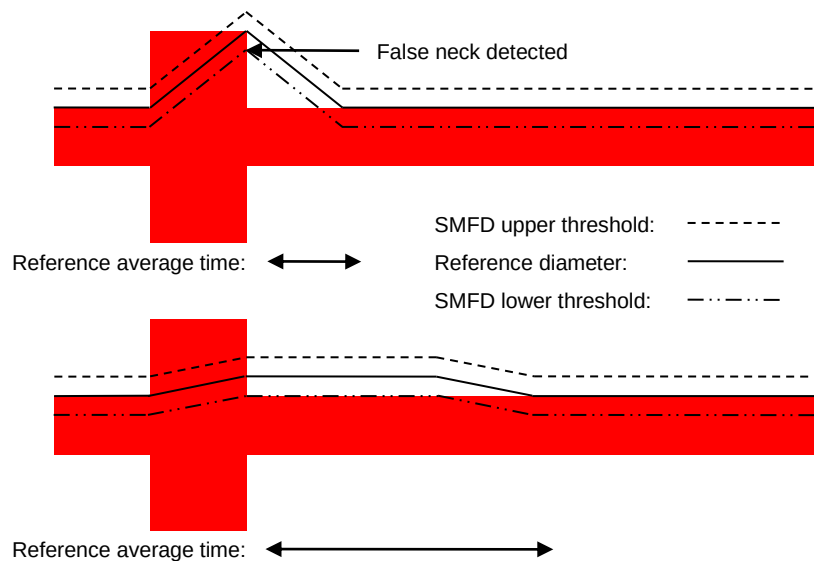
Flaws are detected when the "Measurement Diameter" deviates from the "Reference Diameter" by more than the preset upper or lower SMFD flaw limits.

The "Measurement Average Time" should be set to smooth out any short duration noise on the measurement; it may be left at its default value of 1ms unless measurement noise is large.

The "Reference Average Time" should be set to approximately 100 times the time duration of the longest expected flaw.

If the "Reference Average Time" is set to too short a duration (i.e. close to the time duration of the flaws), then the flaws will significantly affect the value of the "Reference Diameter" and may result in the detection of false flaws.

If the gauge has just been powered on or a cable has just been inserted into the empty optical gate, then the calculated “Reference Diameter” will only be valid after the “Reference Average Time” has elapsed. During this period, multiple false neck flaws will be detected. Hence the “Reference Average Time” should not be set to an excessively large value in order to minimise the ramp up time for the “Reference Diameter”.



Start speed

Setting a minimum “Start speed” prevents the SMFD function from logging flaws during line start-up and shutdown.

Start length

Setting a minimum “Length” prevents the SMFD function from logging flaws during line start-up and shutdown.

COMMUNICATIONS

Comms 1/4

RS232 baudrate: 4800

RS232 mode: Modbus

RS422 baudrate: 4800

RS422 mode: Modbus

CAN Baudrate: 250K

Gauge CAN address: 0

CDI4 CAN address: 0

CDI4 CAN terminator: OFF

⏮ ⏭ ⏭

- Select RS232 baud rate from options: 4800; 9600; 19200; 38400; 115200.
- Select RS232 mode from options: PROTON; Modbus.
- Select RS422 baud rate from options: 4800; 9600; 19200; 38400; 115200.
- Select RS422 mode from options: PROTON; Modbus.
- Select CAN baud rate from options: 250K; 500K; 1000K.
- Set the gauge CAN address.
- Set the CDI4 CAN address.
- Set the CDI4 CAN terminator ON or OFF.

Press the  button to return to the home page.
 Press the  button to access the function menu page.
 Press the  button to advance to the next page.

Comms / Ethernet 2/4

Modbus ID: 0

DHCP: Disabled

IP address: 0 . 0 . 0 . 0

Subnet mask: 0 . 0 . 0 . 0

Gateway: 0 . 0 . 0 . 0

MAC address F8-95-50-01-00-06

UDP IP: 192.168.0. 0

Interval time of UDP[ms]: 0

⏮ ⏭ ⏭

- Set the Modbus ID.
- Enable or disable the DHCP function.
- Set the IP address of the gauge head then click the "OK" button to confirm.
- Set the Subnet Mask of the network then click the "OK" button to confirm.
- Set the Gateway IP address of the network then click the "OK" button to confirm.
- The MAC address of the gauge.
- Set the UDP IP address.
- Set the Interval time of UDP.

Press the  button to return to the home page.
 Press the  button to access the function menu page.
 Press the  button to advance to the next page.

Comms / iBus 3/4

iBUS

iBus type: MODBUS_TCP

iBUS type: NONE

DeviceNet ID: 0

DeviceNet baud rate: 125K

PROFIBUS ID: 0

DHCP: Disabled

Endian mode: Big

Home Menu Advance

- Select the industrial communication bus type.

Note that if the iBUS type is changed, the gauge must be reboot to take effect.

- Set the device ID of the gauge.

- Select the DeviceNet baud rate from options: 125K; 250K; 500K.

- Set the Profibus ID of the gauge then click the “OK” button to confirm.

- Enable or disable the DHCP function.

- Set the Endian mode.

Press the  button to return to the home page.

Press the  button to access the function menu page.

Press the  button to advance to the next page.

Comms / iBus 4/4

iBUS

IP address: 0.0.0.0

Subnet mask: 0.0.0.0

Gateway: 0.0.0.0

MAC address F8-96-50-01-00-01

Home Menu Advance

- Enter the gauge IP address.

- Enter the subnet mask of the network.

- Enter the gateway IP address of the network.

Press the  button to return to the home page.

Press the  button to access the function menu page.

Press the  button to advance to the next page.

INTERFACE

The screenshot shows a black interface with white text. At the top, a header bar contains the word 'Interface' on the left and '1/4' on the right. Below this, there are six rows of settings, each with a label on the left and a selection box on the right. The settings are: 'AOP 1 function:' with 'Average', 'AOP 1 full scale [mm]:' with '0.000', 'AOP 2 function:' with 'Average', 'AOP 2 full scale [mm]:' with '0.000', 'AOP 3 function:' with 'Average', and 'AOP 3 full scale [mm]:' with '0.000'. Below these is 'AOP response:' with 'Instant'. At the bottom, there are three buttons: a home icon, a list icon, and a right arrow icon.

Setting	Value
AOP 1 function:	Average
AOP 1 full scale [mm]:	0.000
AOP 2 function:	Average
AOP 2 full scale [mm]:	0.000
AOP 3 function:	Average
AOP 3 full scale [mm]:	0.000
AOP response:	Instant

Press the **Interface** icon on the function menu page to access the Interface page 1.

- Select the analogue output 1 function.

- Set the analogue output 1 full scale.

- Select the analogue output 2 function.

- Set the analogue output 2 full scale.

- Select the analogue output 3 function.

- Set the analogue output 3 full scale.

- Select the analogue output response .

Press the  button to return to the home page.

Press the  button to access the function menu page.

Press the  button to advance to the next page.

AOP 1, 2, 3 function

Select the analogue output 1, 2, 3 function from options:

Average;

X;

Y;

Z;

Ovality;

Average error;

X error; Y error;

Z error;

Ovality error.

AOP 1, 2, 3 full scale

Set the maximum measured diameter or most positive error for a +10V output voltage.

- For unipolar outputs, a zero measured diameter corresponds to a 0V output voltage.
- For bipolar outputs, the most negative measured error corresponds to a -10V output voltage.

AOP response: Instant; Average.

Instant: Analogue output voltage represents the value of each individual measurement.

Average: Analogue output voltage represents the time averaged measured value.

Interface
2/4

Relay 1:
Gauge OK

Relay 2:
Gauge OK

Relay 3:
Gauge OK

Relay 4:
Gauge OK

Closure time [ms]:
0





- Select the condition under which this relay contact will close (short-circuit).

- Same as above.

- Same as above.

- Same as above.

Set the relay contact closure time duration (this setting applies to all four relays).

Press the  button to return to the home page.

Press the  button to access the function menu page.

Press the  button to advance to the next page.

Relay 1, 2, 3, 4

Select form the following options:

Gauge OK;
 Over any (limit);
 Under any (limit);
 Lump&Neck;
 Lump;
 Neck;
 Over average;
 Under average;
 Over X;
 Under X;
 Over Y;
 Under Y;
 Over Z;
 Under Z;
 Over ovality;
 Under ovality.

Interface

3/4

LIN1 function:

Reset

LIN1 polarity:

Active low

LIN2 function:

Reset

LIN2 polarity:

Active low

Ext-alarm hold [ms]:

0

Speed source:

Preset

Preset speed [m/min]:

0

>>

- Select the logic input 1 function from options: Reset; Ext alarm 1; Ext alarm 2; Line rev direction.

- Select the logic input 1 polarity from options: Active low; Active high.

- Select the logic input 2 function from options: Reset; Ext alarm 1; Ext alarm 2; Line rev direction.

- Select the logic input 2 polarity from options: Active low; Active high.

- Set the Ext-alarm hold time.

- Select the speed source from options: Preset; Pulse; Analogue.

- Set the preset speed (only when “Speed source” above is set to “Preset”).

Press the  button to return to the home page.

Press the  button to access the function menu page.

Press the  button to advance to the next page.

LIN 1, 2, function

Reset:

- Reset all flaw sizes, counts and positions to zero.
- Reset all running maximum and minimum diameters to zero.
- Reset cumulative length to zero.

Ext alarm 1:

- Print “Ext alarm 1” message to a printer connected to the RS232 port.
- Set output parameter bit DW1.8=1

Ext alarm 2:

- Print “Ext alarm 2” message to a printer connected to the RS-232 port.
- Set output parameter bit DW1.9=1.

Line rev direction: Decrement the length of the line.

LIN 1, 2 polarity

Active low: Active when applied input voltage is below +3V.

Active high: Active when applied input voltage is above +10.5V.

Ext-alarm hold

Set a minimum hold time (in milliseconds) for the “Ext alarm 1” and “Ext alarm 2” logic inputs; multiple activations of any “Ext alarm 1” and “Ext alarm 2” logic input within this time period are counted as a single activation.

Speed source

Preset: Use the fixed “Preset speed” value for the line speed.

Pulse: Determine the line speed from the pulse input.

Analogue: Determine the line speed from the analogue input voltage.

Interface / PID 4/4

PID range [%]:

Distance [m]:

Extruder delay [s]:

Start speed [m/min]:

PID P [%]:

PID I [%]:

PID output polarity:

- Set the PID controller output voltage range (0 to 50%).

- Set the wire path distance between the gauge and the PID controlled equipment (extruder or capstan).

- Set the response time of the PID controlled equipment (extruder or capstan; typically 1s).

- Set the minimum line speed for PID controller operation.

- Set the diameter error proportional gain (0 to 100%).

- Set the diameter error integral gain (0 to 100%).

- Select the PID controller output polarity from options: Normal; Reverse.

Press the  button to return to the home page.

Press the  button to access the function menu page.

Press the  button to advance to the next page.

Gauge Information

Gauge Info 1/1

Gauge hardware ver:

Gauge core ver:

Gauge app Ver:

Gauge date:

CDI4 software ver:

CDI4 software date:

Press the **Gauge Info** icon on the function menu page to access this page.

- Gauge hardware version.

- Gauge core version.

- Gauge application version.

- Gauge firmware issue date.

- CDI4 software version.

- CDI4 software issue date.

Press the  button to return to the home page.

Press the  button to access the function menu page.

Store



Press the **Store** icon on the function menu page to access this page.

- Enter the parameter group (0 to 49).

- Click the **“Recall”** button to recall the stored input parameter values in the group.

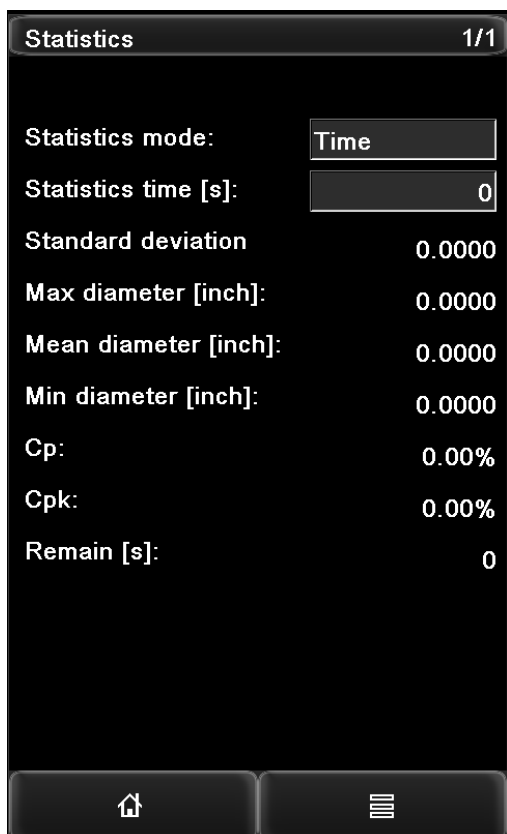
- Click the **“Save”** button to save changes to the current active group after modifying the input parameter values.

Note: When the Group authority was set to “Locked” on the Access level page, only “Recall” operation can be executed.

Press the  button to return to the home page.

Press the  button to access the function menu page.

Statistics



Press the **Statistics** icon on the function menu page to access this page.

- Set the statistics mode from options: Time; Length; One Reel.

- Set the statistics time (if “Statistics mode” is set to “Time”) or length (if “Statistics mode” is set to “Length”).

- Standard deviation of all diameters sampled during the statistics.

- Maximum diameter sampled during the statistics time.

- Mean of all diameters sampled during the statistics time.

- Minimum diameter sampled during the statistics time.

- Process capability Cp

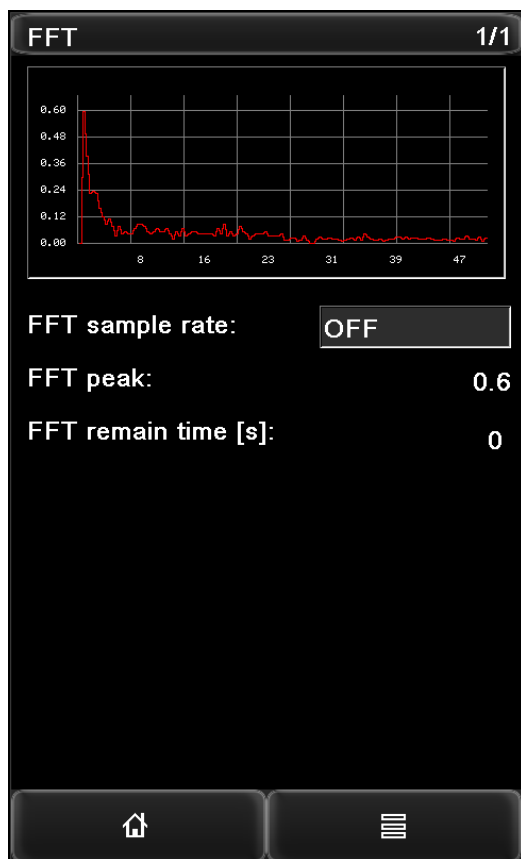
- Process capability Cpk

- Count down of the remaining statistics time or length until the next statistics result.

Press the  button to return to the home page.

Press the  button to access the function menu page.

FFT



Press the **FFT** icon on the function menu page to access this page.

- FFT graph.

- Select the FFT sampling rate from options: 1Hz; 3Hz; 10Hz; 30Hz; 100Hz; 300Hz; 1KHz; or select "OFF" to close the FFT function.

- FFT peak value.

- FFT sampling window remaining time.

Press the  button to return to the home page.

Press the  button to access the function menu page.

ACCESS LEVELS

Press the **Access levels** icon on the function menu page and input the password “18018” to enter the access levels pages.

Access levels 1/2	
Home Pages:	Unlocked
Preset pages:	Locked
Measurement pages:	Unlocked
Comms pages:	Unlocked
Interface pages:	Unlocked
Store pages:	Unlocked
Statistics pages:	Unlocked
FFT pages:	Unlocked

Access levels 2/2	
Page 2 of Home visible:	TRUE
Page 3 of Home visible:	TRUE
Page 4 of Home visible:	TRUE
Page 5 of Home visible:	TRUE

The “Access levels” page is a password protected page which restricts the setting of parameters.

The function menu pages can be locked or unlocked by selecting “Locked” or “Unlocked”.

The Home pages 2, 3, 4, 5 can be set to be displayed or hidden by selecting “TRUE” or “FALSE”.

It is necessary to unlock the relevant pages before their parameters can be set via the CDI4 interface.

STANDARD COMMUNICATIONS INTERFACES

CAN-BUS COMMUNICATIONS

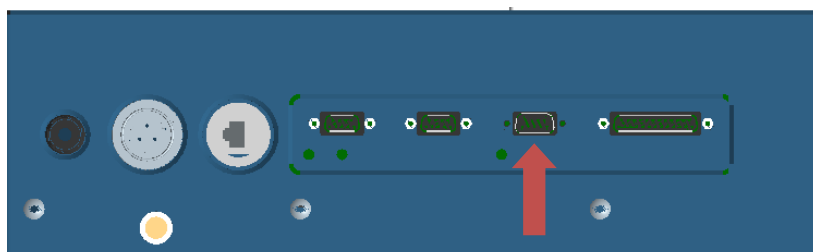
This CAN-bus interface is fitted as standard. It operates independently of the other communications interfaces and may be accessed at the same time as them.

The CAN-bus interface uses a proprietary Proton Products protocol. It is exclusively used to communicate between the unit and other Proton Products modules. The unit automatically detects connection to other modules and configures the bus appropriately; it is not normally necessary to manually configure this interface.

The CAN-bus interface is not intended for use with an external CAN-bus network.

CAN-bus interface

The CAN-bus interface may be accessed through the “CAN / POWER” connector.



Connector type: DB9 male (plug)

Pin	Designation	Comment
2	CANL	
3	GND	Ground reference (isolated from earth / shield).
5	Shield	Ensure that the cable shield is connected to this via the plug shield connection.
7	CANH	
9	+24V	+24VDC power input (the gauge may be powered via this pin).
S	Shield	Ensure that the cable shield is connected to the plug shield connection.

CAN-bus LED indicator

CAN		LED status	Indication
		Flashing green	Online
CAN-bus communications		Flashing red	Communication error
		Extinguished	No communication

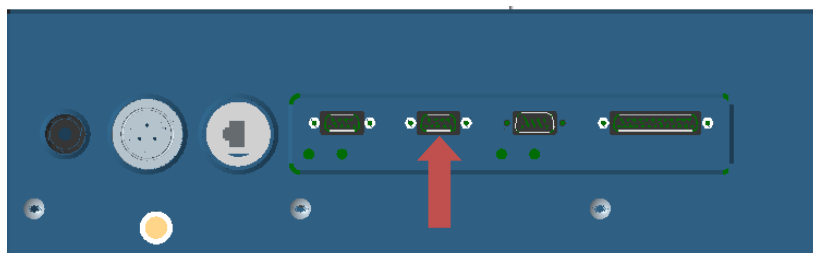
RS-232 COMMUNICATIONS

This RS-232 interface is fitted as standard. It operates independently of the other communications interfaces and may be accessed at the same time as them.

If the RS-232 interface “Mode” is set to “Print”, then an “Epson TM-U220D” or compatible printer may be connected to the port and configured to print fault and reel report tickets.

RS-232 interface

The RS-232 interface may be accessed through the “RS232/485” connector.



Connector type: DB9 female (socket)

Pin	Designation	Comment	PC DB9 serial port pin
2	TXD1		2
3	RXD1		3
5	GND_R	Ground reference is not isolated from earth.	5
7	CTS1		Not used
8	RTS1		Not used
S	Shield	Ensure that the cable shield is connected to this via the plug shield connection.	Shield

The above table also shows the configuration of a cable for connection to a personal computer (PC) type DB9 serial port.

The maximum baud rate depends on the cable capacitance and length. For low-cost overall shielded cable with total capacitance of shield to core-plus-core to core of 300pF per metre, the maximum recommended baud rates are as follows:

Cable length range / m		Maximum Baud rate / s
0	3	115200
3	10	38400
10	20	19200
20	40	9600
40	80	4800

RS-232 SINGLE LETTER PROTOCOL (SLP)

The DG-k series diameter gauges support the legacy Single Letter Protocol (SLP) for communications over the RS-232 interface.

The SLP RS-232 data format is:

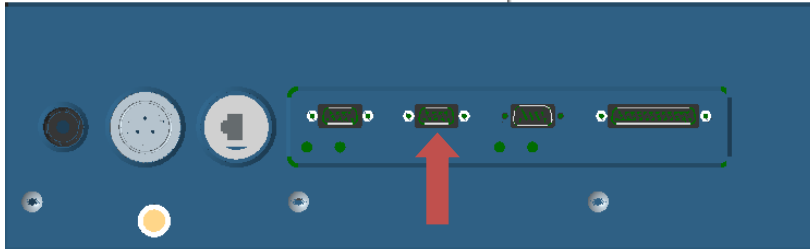
Number of data bits	Parity	Number of stop bits	Flow control	Default baud rate
7	None	2	None	115200 bit / s

RS-422 / RS-485 COMMUNICATIONS

This RS-422 / RS-485 interface is fitted as standard. It operates independently of the other communications interfaces and may be accessed at the same time as them.

RS-422 / RS-485 interface

The RS-422 / RS-485 interface may be accessed through the “RS232/485” connector.



Connector type: DB9 female (socket)

Pin	RS-422 Designation	Comment	RS-485 two wire operation		
			Link	Link	Designation
1	RS422_A		●		A+
4	RS422_B		●	●	B-
5	GND_R		●	●	GND
6	RS422_Y		●		
9	RS422_Z			●	
S	Shield	Ensure that the cable shield is connected to this via the plug shield connection.			Shield

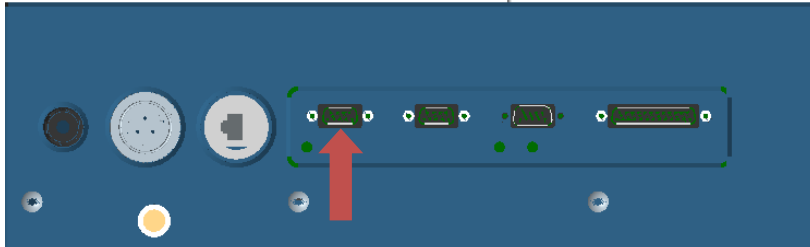
The four-wire RS-422 interface may be converted to two-wire RS-485 operation by installation of the 2 shorting links shown above.

PROFIBUS COMMUNICATIONS

This communications interface operates independently of the other communications interfaces and may be used at the same time as them.

PROFIBUS interface

The PROFIBUS interface may be accessed through the “PROFI/EIP/DEV” connector.



Connector type: DB9 female (socket)

Pin	PROFIBUS	Comments
3	B	
4	RTS	
5	GND	
6	+5V	
8	A	
S	Shield	Ensure that the cable shield is connected to this via the plug shield connection.

PROFIBUS LED indicator

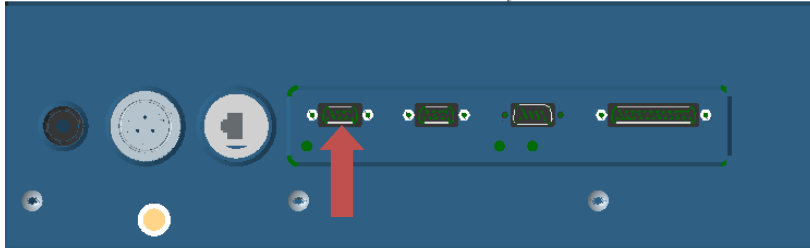
i-BUS		LED status	Indication
		Continuous green	Online
PROFIBUS communications		Continuous red	Communication error
		Extinguished	No communication

PROFINET communications

This communications interface operates independently of the other communications interfaces and may be used at the same time as them.

PROFINET interface

The PROFINET interface may be accessed through the “PROFI/EIP/DEV” connector.



Connector type: DB9 female (socket)

Pin	PROFINET	Comments
1	LAN TX-	
2	LAN TX+	
7	LAN RX-	
9	LAN RX+	
S	Shield	Ensure that the cable shield is connected to this via the plug shield connection.

PROFINET LED indicator

i-BUS		LED status	Indication
		Continuous green	Online
PROFINET communications		Continuous red	Communication error
		Extinguished	No communication

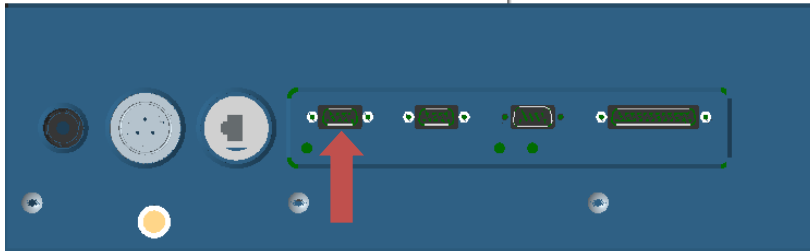
ETHERNET / IP COMMUNICATIONS

The Ethernet/IP interface is compatible with PLCs fitted with an Ethernet/IP interface that supports the CIP (Common Industrial Protocol).

This communications interface operates independently of the other communications interfaces and may be used at the same time as them.

EtherNet / IP interface

The EtherNet / IP interface may be accessed through the “PROFI/EIP/DEV” connector.



Connector type: DB9 female (socket)

Pin	EtherNet/IP	Comments
1	LAN TX-	
2	LAN TX+	
7	LAN RX-	
9	LAN RX+	
S	Shield	Ensure that the cable shield is connected to this via the plug shield connection.

EtherNet / IP LED indicator

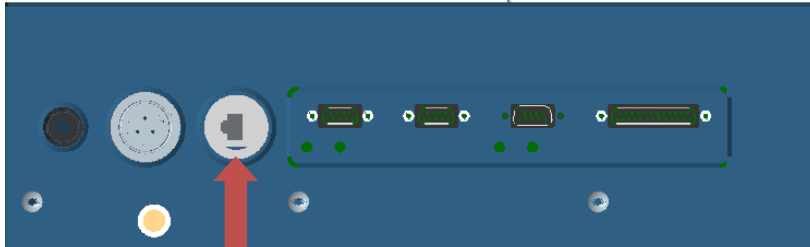
i-BUS		LED status	Indication
		Continuous green	Online
EtherNet / IP communications		Continuous red	Communication error
		Extinguished	No communication

ETHERNET COMMUNICATIONS

This Ethernet interface is fitted as standard. It operates independently of the other communications interfaces and may be accessed at the same time as them.

Ethernet interface

The Ethernet interface may be accessed through the RJ45 connector.



Connector type: RJ45 8P8C female (socket)

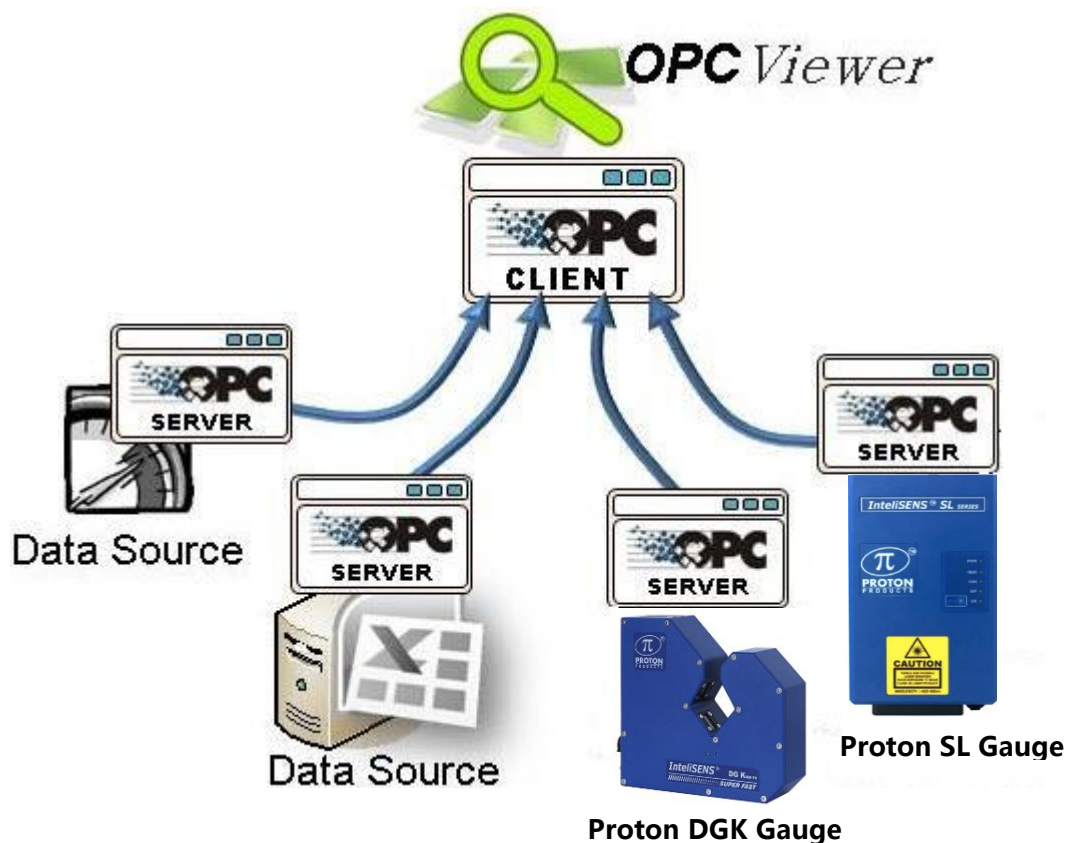
Pin	Designation	Comments
1	LAN TX+	
2	LAN TX-	
3	LAN RX+	
6	LAN RX-	
S	Shield	Ensure that the cable shield is connected to this via the plug shield connection.

Ethernet LED indicator

EIP		LED status	Indication
		Continuous green	Online and connected
Ethernet communications		Flashing green	Online but not connected
		Continuous red	Critical link failure
		Flashing red	Connection timeout

OPC UNIFIED ARCHITECTURE (UA)

OPC UA is a central element of Industry 4.0 providing seamless communication between IT and industrial production systems. OPC UA 1.03 has been supported on Proton DGK gauge as a standard interface. The OPC Client can exchange data (read / write) with DGK gauge (OPC Server) to real-time monitor and manage the gauge.

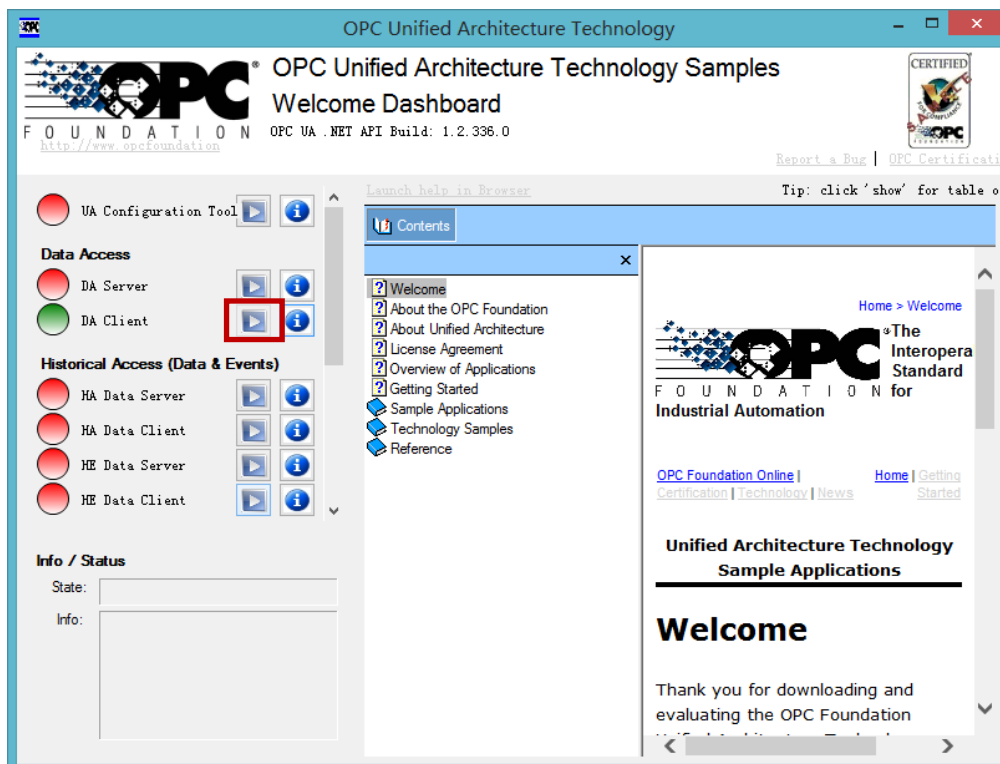


To connect to the DGK gauge (OPC Server) from an OPC Client software, please use the following procedure:

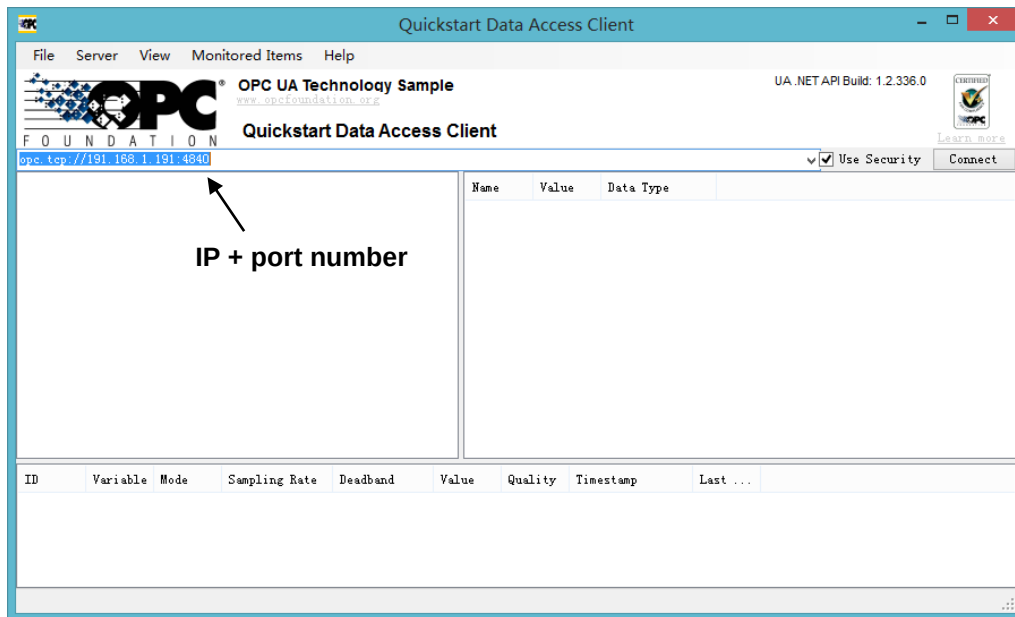
1. Set up the OPC port number, user name and password on PCIS.
Main page > Comms (password: 65000) > OPC

Enter the OPC port number (default: 4840), user name (default: admin) and password (default: admin). If “Anonymous” is ticked, the OPC Client will log in the gauge anonymously.

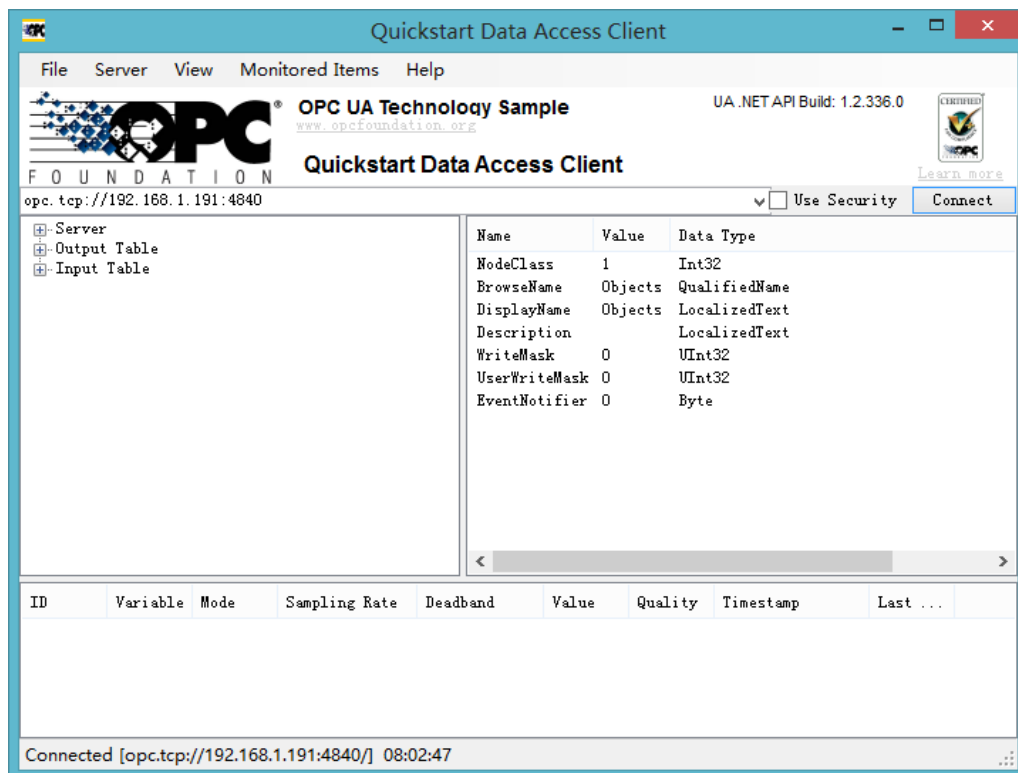
2. Configuration on the OPC Client (take OPC UA Dashboard” as an example. It’s similar for other OPC Client software). Note that only one OPC Client can connect to the gauge at the same time.



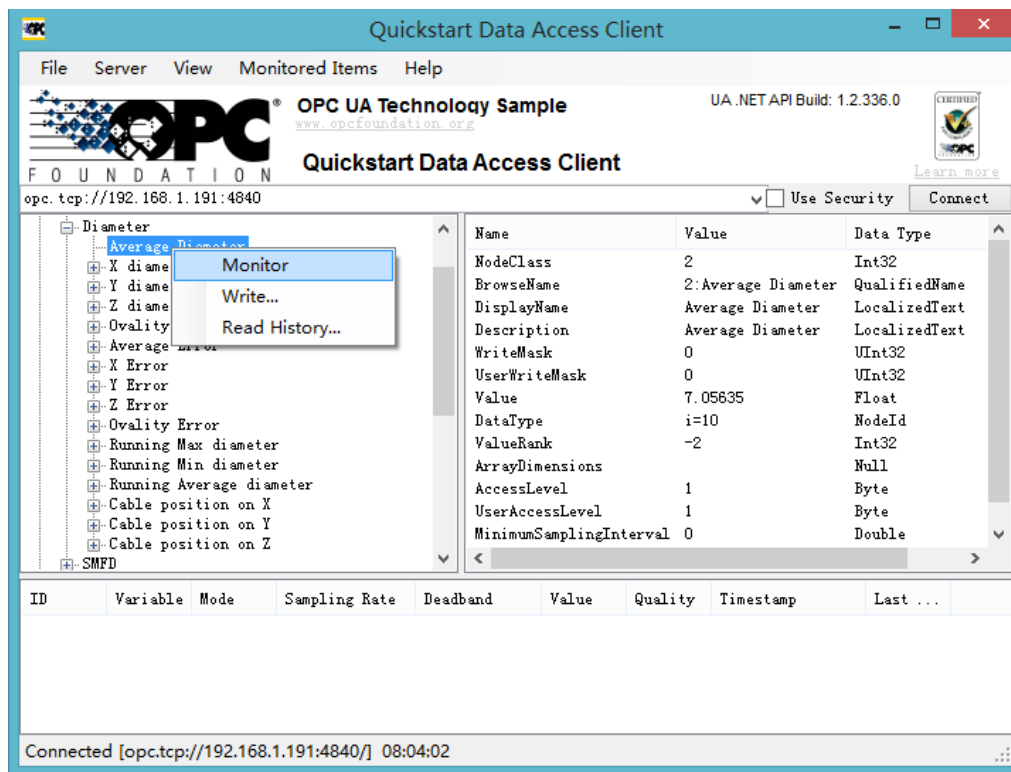
Click “DA Client” button to enter the Data Access page.



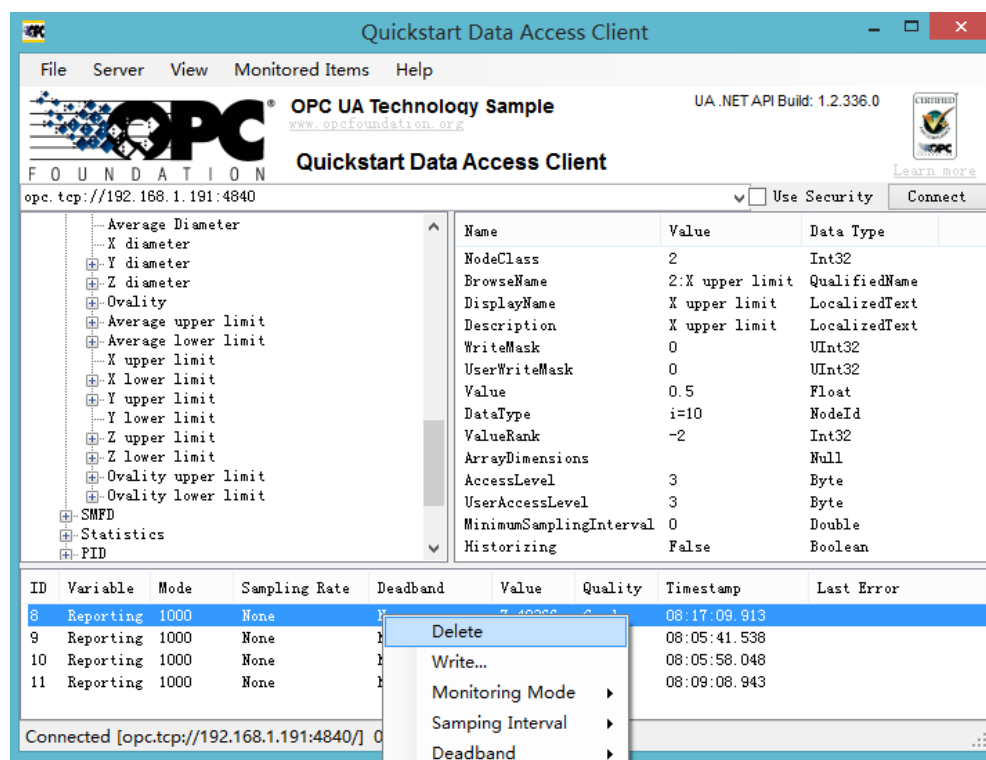
Enter the gauge IP address and port number then click “Connect” button. Note that the IP address should be the same as the one configured on PCIS.



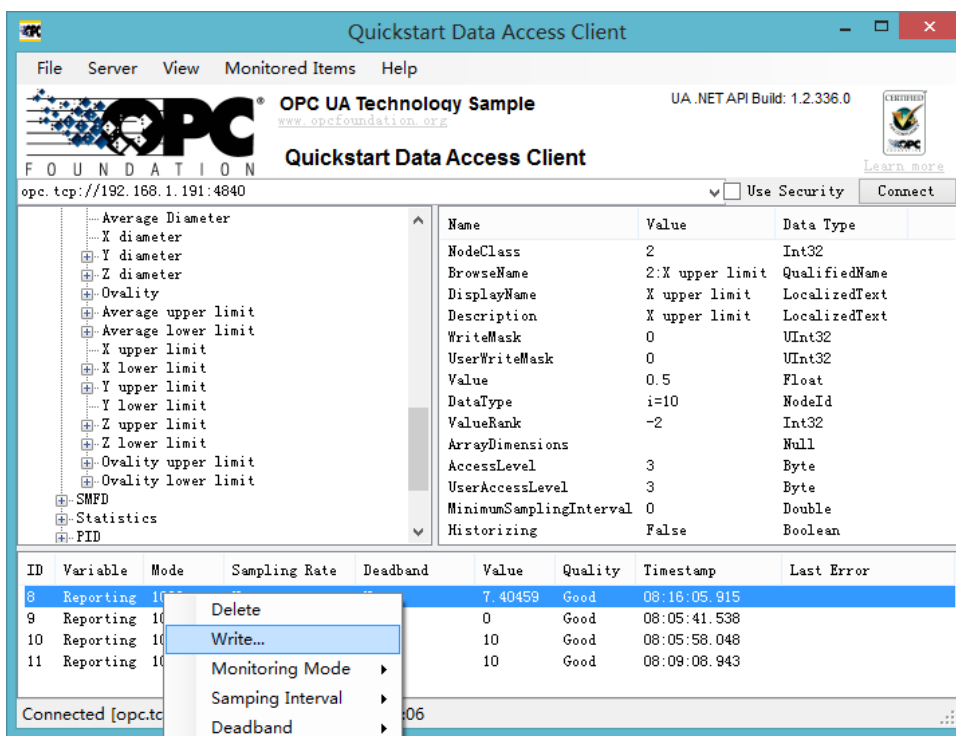
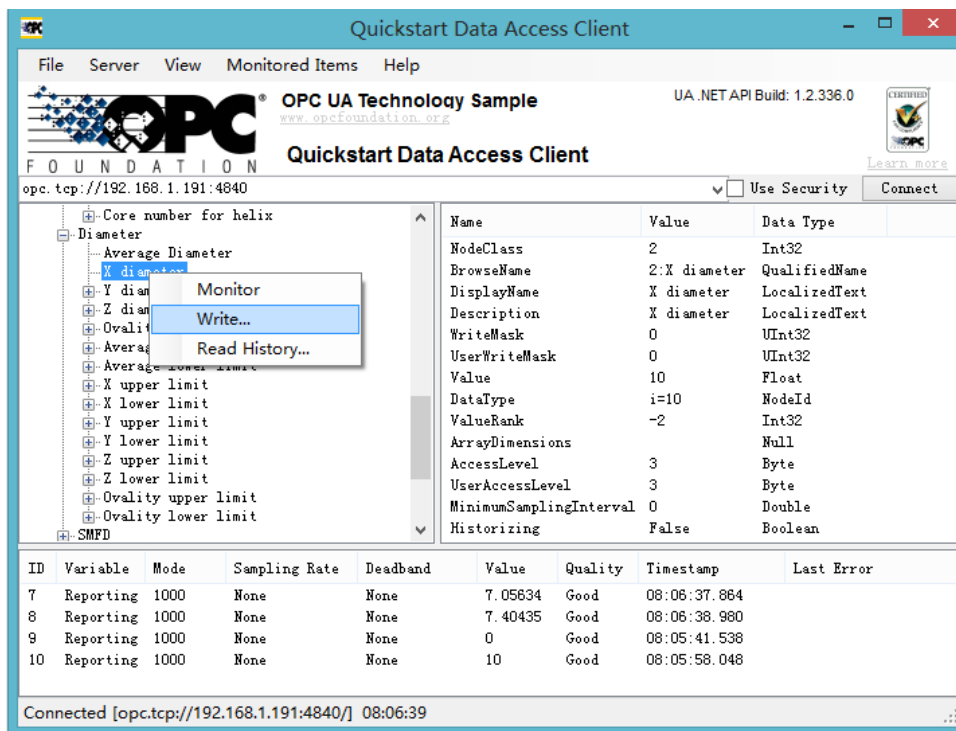
Unfold the parameter list on the left side of the page, right-click and select **Monitor** on the pop-up menu, for example, Average diameter, the parameter will appear in the monitoring list



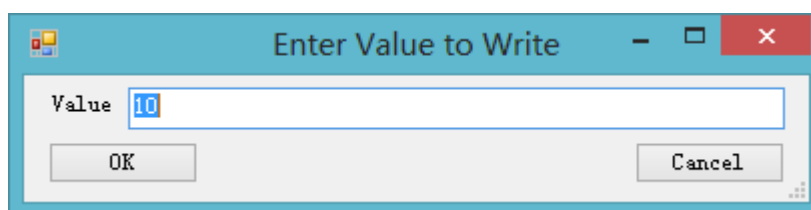
Right-click on the monitored parameter and select **Delete** on the pop-up menu to stop monitoring the parameter.



Right-click and select **Write** on the pop-up menu to write a parameter to the gauge.



Enter the parameter value on the pop-up menu then click “OK” button to confirm.



STANDARD ELECTRICAL INTERFACES

SPEED PULSE INPUT

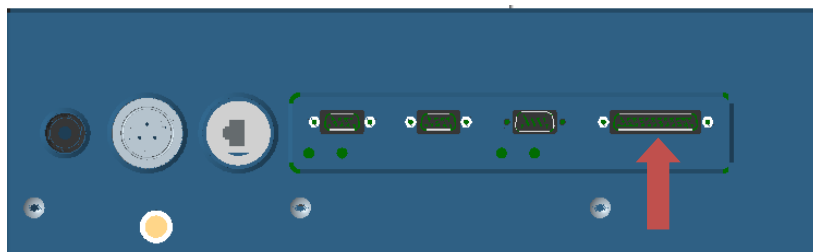
The speed pulse input may be connected to a line speed gauge (such as a Proton Products SL or SLR series non-contact laser speed and length gauge).

Measured line speed input is required for:

- Helix mode operation.
- PI feedback controller operation.
- Statistical Process Control (SPC) operation.

Speed pulse input connection

The speed pulse input is fitted as standard and may be accessed through the “INTERFACE” connector.



Connector type: DB25 female (socket)

Pin	Designation	Description	Notes
11	DGND	Digital ground	Ground reference for SPD1 and SPD2.
12	SPD2	Speed pulse input 2	For low voltage speed pulses (e.g. 5V TTL).
13	SPD1	Speed pulse input 1	For high voltage speed pulses (e.g. 12 ~ 24V speed encoders).
S	Shield		Ensure that the cable shield is connected to this via the plug shield connection.

Speed pulse input electrical specification

- A choice of two speed pulse inputs are available: one input is for high voltage pulses and one input is for low voltage pulses; only one input should be selected for use.
- The ground reference (GND) for the speed pulse inputs is not isolated from earth.

Specification		Min	Typ	Max	Units
SPD1 (pin13) High voltage (e.g. 12 ~ 24V speed encoders)	Low state (logic 0) input voltage			+5	V
	High state (logic 1) input voltage	+9			V
	Absolute input voltage	-50		+50	V
SPD2 (pin 12) Low voltage (e.g. 5V TTL)	Low state (logic 0) input voltage			+2	V
	High state (logic 1) input voltage	+4			V
	Absolute input voltage	-30		+30	V
Pulse frequency				250	kHz

ANALOGUE INPUT

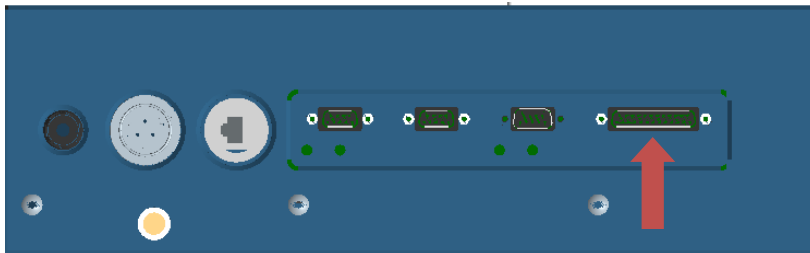
The analogue input may be connected to a 0 to +10V signal from a line speed gauge.

Measured line speed input is required for:

- Helix mode operation.
- PI feedback controller operation.
- Statistical Process Control (SPC) operation.

Analogue input connection

If installed, the analogue input may be accessed through the “INTERFACE” connector.



Connector type: DB25 female (socket)

Pin	Designation	Description	Notes
9	AGND	Analogue input ground	Isolated ground reference for AIP
22	AIP	Analogue input	
S	Shield		Ensure that the cable shield is connected to this via the plug shield connection.

Analogue input electrical specifications

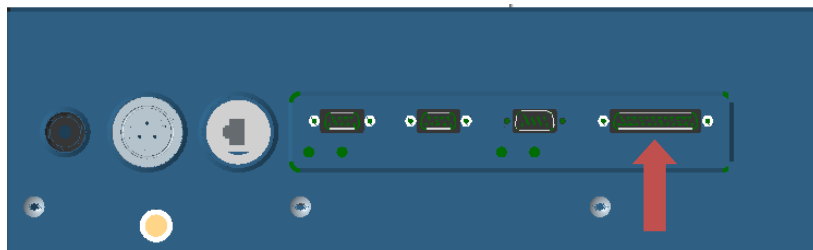
- The analogue input and its ground reference are isolated from earth.

Specification	Minimum	Typical	Maximum	Units
Input voltage measurement range	0		+10	V
Input voltage absolute range	-35		+35	
Input-to-earth voltage	-50		+50	V
Input resistance	8			k Ω

LOGIC INPUTS

Logic inputs connection

Two logic inputs are fitted as standard and may be accessed through the “INTERFACE” connector.



Connector type: DB25 female (socket)

Pin	Designation	Description	Notes
10	LIN2	Logic input 2	
11	DGND	Digital ground	Ground reference for LIN1 and LIN2.
23	LIN1	Logic input 1	
S	Shield		Ensure that the cable shield is connected to this via the plug shield connection.

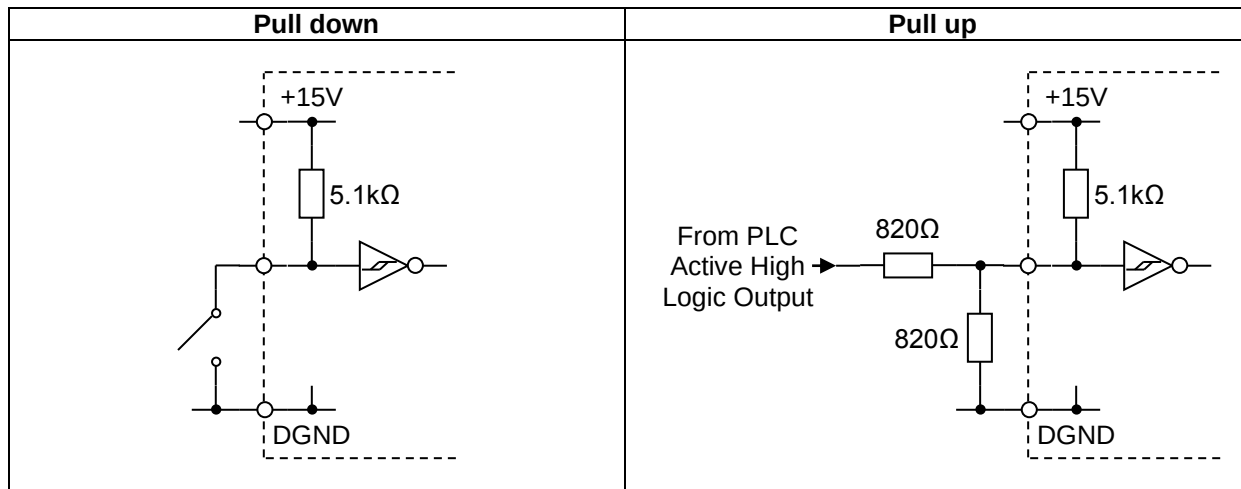
Logic inputs electrical specification

- Logic inputs are not isolated from earth.
- Inputs are internally pulled up to +15 V via 5.1k Ω resistors; unconnected inputs default to the high state.
- Inputs will source a minimum current of 3mA when externally pulled down to the low state.

Specification	Minimum	Typical	Maximum	Units
Low state (logic 0) input voltage			3	V
High state (logic 1) input voltage	10.5			V
Absolute input voltage	-30		30	V
Low state source current	3			mA

Logic inputs can be connected to different types of signal sources as follows:

Signal source	Connection method
Mechanical or solid state relay contact	Connect between logic input and DGND (pin 11).
Pull-down	Connect the pull-down signal to the logic input and ground to DGND (pin 11).
Pull-up	• Connect an 820Ω resistor between the logic input and DGND (pin 11) to pull the input down to 2.5V. • Connect the pull-up signal source to the logic input and ground to DGND (pin 11). • For 24V signals (such as from a PLC), the power dissipation in the 820Ω resistor can be reduced by connecting another 820Ω in series between the pull-up signal source and logic input.



Logic inputs configuration

Each individual logic input can be independently configured for both active state polarity and function:

Function bit setting	Logic input function		
	Function	Inactive	Active
0	Reset	-	Reset all flaw detection measurements to zero.
			Reset all running maximum and minimum diameters to zero.
			Reset length to zero.
1	Ext Alarm 1	Set output parameter bit DW1.8 = 0	Print “ EXT-ALARM-1 ” message to a printer connected to the RS-232 port.
			Set output parameter bit DW1.8 = 1
2	Ext Alarm 2	Set output parameter bit DW1.9 = 0	Print “ EXT-ALARM-2 ” message to a printer connected to the RS-232 port.
			Set output parameter bit DW1.9 = 1
3	Line Rev Direction		Decrement the length of the line.

Input DW logic input polarity bit setting	Logic input active state
0	Active low
1	Active high

Interrogating External Alarm 1 and External Alarm 2 Logic Input Status

The status of logic inputs assigned with the “External Alarm 1” and “External Alarm 2” functions can be interrogated via output parameter bits DW1.8 and DW1.9 respectively:

Output parameters

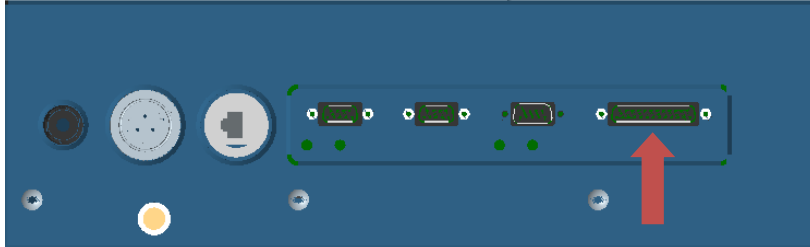
DW	Bit	Comments	Units	Range/Remark
1	0	Res		
	1	No reading		0=OK 1=No Reading
	2	No object		0=OK 1=No Object
	3	Window dirty		0=OK 1=Window Dirty
	4	Line speed too low for helix mode		0=OK 1=Too Low

	5	Line speed too high for helix mode		0=OK 1=Too High
	6	Gauge overheat		0=OK 1=Gauge overheat
	7			
	8	External alarm 1		0=OK 1=Alarm
	9	External alarm 2		0=OK 1=Alarm
	10~ 15			

RELAY OUTPUTS

Relay outputs connection

Four relay outputs are fitted as standard and may be accessed through the “INTERFACE” connector.



Connector type: DB25 female (socket)

Pin	Designation	Description	Notes
17	NO1	Relay 1 normally open contact	
5	COM1	Relay 1 common	
18	NO2	Relay 2 normally open contact	
6	COM2	Relay 2 common	
19	NO3	Relay 3 normally open contact	
7	COM3	Relay 3 common	
20	NO4	Relay 4 normally open contact	
8	COM4	Relay 4 common	
S	Shield	Shield	Ensure that the cable shield is connected to this via the plug shield connection.

Relay outputs electrical specifications

- All contacts are isolated and voltage free.
- If the relay contacts are used to switch inductive loads, then suppressors must be fitted to clamp voltage spikes.

Specification	Minimum	Typical	Maximum	Units
DC voltage between contacts			24	VDC
Contact-to-earth voltage			50	V
Current			1	A
On resistance (at a current of 10mA)			0.1	Ω

OPTIONAL ELECTRICAL INTERFACES

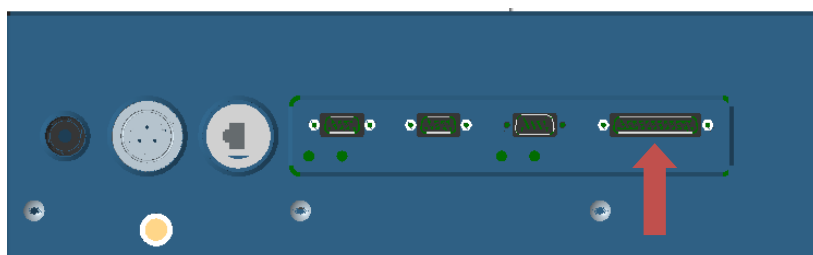
ANALOGUE OUTPUTS

Analogue outputs are an optional extra that must be ordered for installation during manufacture; they cannot be retrofitted to the gauge.

The analogue outputs may be configured to output diameter or error measurements as an analogue output voltage.

Analogue outputs connection

Three analogue outputs may be accessed through the “INTERFACE” connector.



Connector type: DB25 female (socket)

Pin	Designation	Description	Notes
3	AOUT1	Analogue output 1	
4	AOUT3	Analogue output 3	
15	Agnd	Analogue output ground	Isolated ground reference for AOUT1, AOUT2 and AOUT3.
16	AOUT2	Analogue output 2	
S	Shield		Ensure that the cable shield is connected to this via the plug shield connection.

Analogue outputs electrical specifications

- Analogue outputs and their shared ground reference are isolated from earth.
- Analogue outputs are short circuit proof.

Specification	Minimum	Typical	Maximum	Units
Output voltage	-10		10	V
Output-to-earth voltage			50	V
Load resistance	3.3			kΩ

PI FEEDBACK CONTROLLER

The PI feedback controller is an optional extra that must be ordered for installation during manufacture; it cannot be retrofitted to the gauge.

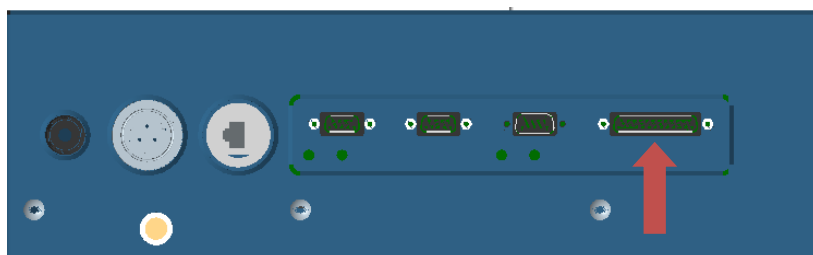
The DG gauge can output an analogue control signal based on the proportional and integral value of the difference between the measured diameter and the $\varnothing$ preset diameter. This signal can be used to control an insulation extruder or capstan driver with $\varnothing$ preset as the target diameter of the product.

For PI feedback controller operation, the gauge requires the following prerequisites:

- A calibrated, real time line speed measurement provided to the gauge either as an analogue or pulse input signal.
- $\varnothing$ preset and $\varnothing$ core diameters set in the Basic $\varnothing$ Data page.

PI feedback controller connection

The PI feedback controller may be accessed through the “INTERFACE” connector.



Connector type: DB25 female (socket)

Pin	Designation	Description	Notes
1	P0V	PI feedback controller ground reference	Isolated from earth.
2	POUT	PI feedback controller output	
14	PIN	PI feedback controller input	
S	Shield		Ensure that the cable shield is connected to this via the plug shield connection.

PI feedback controller electrical specifications

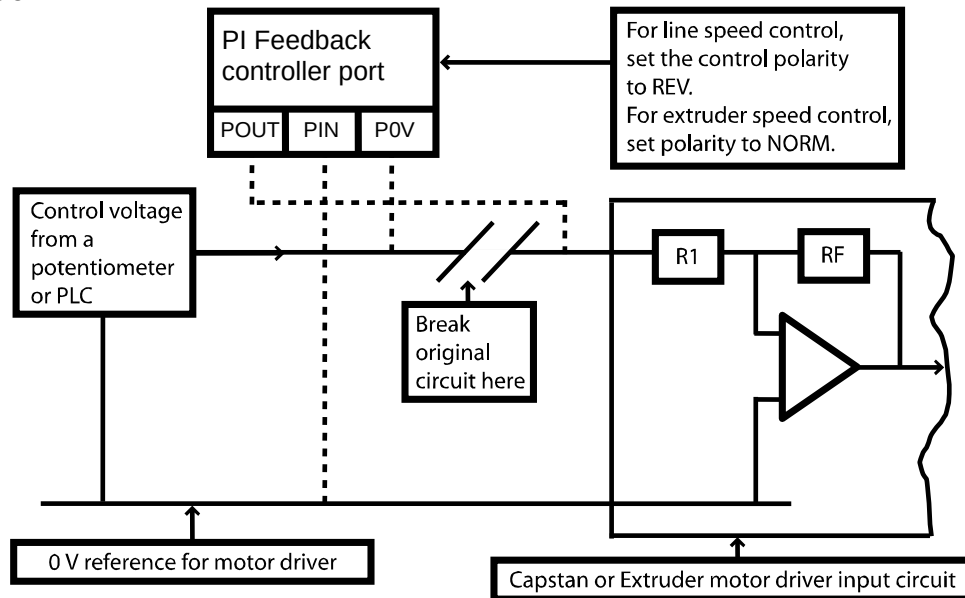
- The ground reference for the PI feedback controller input (PIN) and output (POUT) is isolated from earth.
- POUT is short circuit proof.
- The PI feedback controller output (POUT) voltage range is limited to a user configurable maximum of 50 % of the PI feedback controller input (PIN) voltage range.

Specification	Minimum	Typical	Maximum	Units
PIN input voltage	-20		+20	V
PIN input voltage (absolute)	-50		+50	V
PIN input to earth voltage	-50		+50	V
PIN input impedance	20			k Ω
POUT output voltage	-10		+10	V
POUT output voltage (absolute forced)	-25		+25	V
POUT output to earth voltage	-50		+50	V
POUT output current			10	mA
POUT output impedance			10	Ω

PI feedback controller connection to production equipment

The connection of the PI feedback controller depends on the configuration of the insulation extruder or capstan drive control inputs.

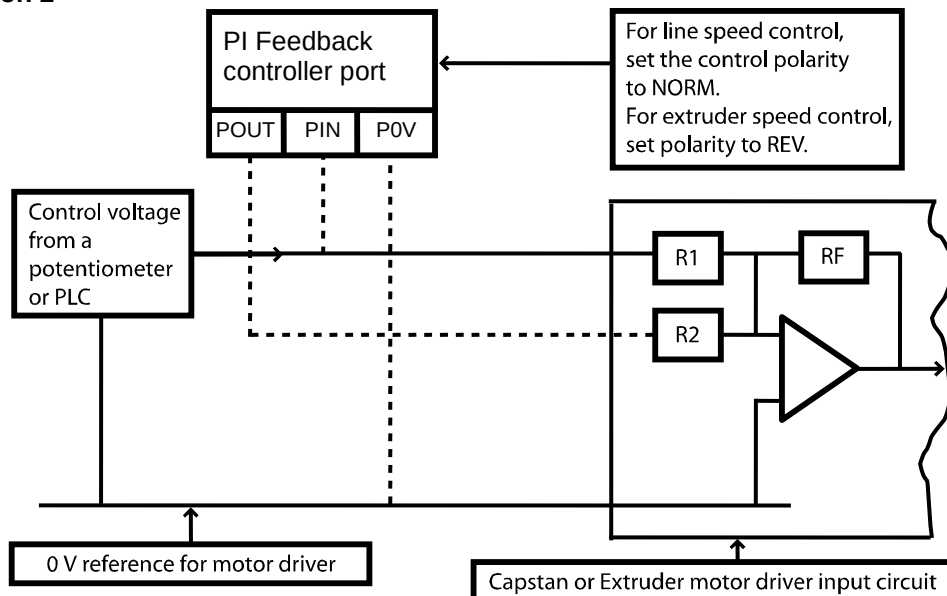
Connection 1



In this configuration, the PI feedback controller output voltage is added in series with a fixed control voltage from a potentiometer or PLC. The fixed control voltage sets the capstan or extruder operating midpoint and the PI feedback controller output drives the capstan or extruder above or below this operating midpoint.

Note that the PI feedback controller ground reference (0V) is not connected to the 0V reference of the controlled equipment; this is possible as the PI feedback controller ground reference is isolated from earth.

Connection 2

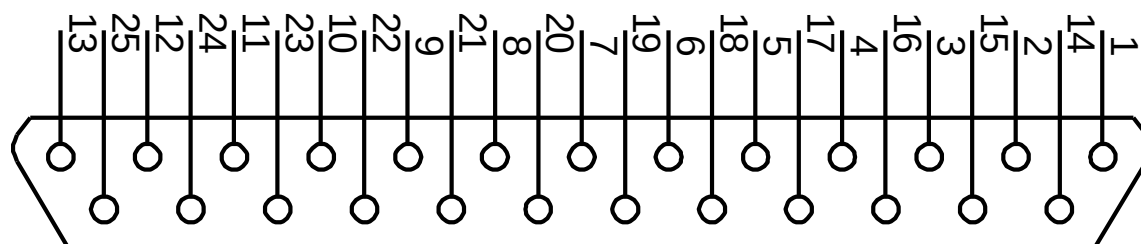
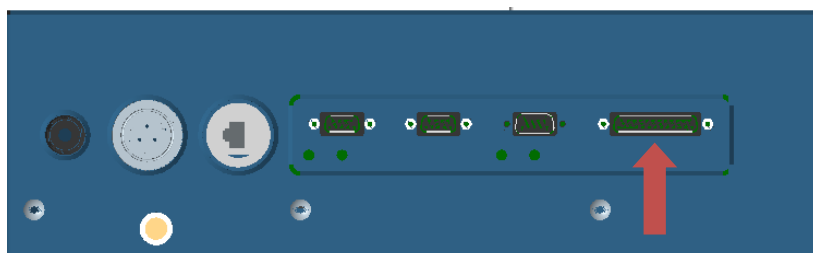


In this configuration, the PI feedback controller output voltage is added to a fixed control voltage from a potentiometer or PLC via the auxiliary control input on the extruder or capstan driver. The fixed

control voltage sets the capstan or extruder operating midpoint and the PI feedback controller output drives the capstan or extruder above or below this operating midpoint.

Note that the PI feedback controller output connects to an auxiliary control input (through R2) with the same scaling as the primary control input (through R1) of the controlled equipment (i.e. $R1 = R2$). If the scaling between the two control inputs is different, then the PI feedback controller gain must be adjusted accordingly. For example, if $R2 = 2 \times R1$, then the gain of the auxiliary control input is half of the primary control input, so the PI feedback controller gain setting must be doubled to compensate and the control range must also be similarly increased. The system cannot function if $R2 > 5 \times R1$ since the maximum control range will be $< 10\%$, which is normally not sufficient.

INTERFACE CONNECTOR PIN OUTS



Connector type: DB25 female (socket)

Pin	Designation	Functional group	Description	Notes
1	P0V	PI feedback controller	PI feedback controller ground	Isolated ground
2	POUT	PI feedback controller	PI feedback controller output	±10V isolated output
3	AOUT1	Analogue outputs	Analogue output 1	±10V full scale isolated output
4	AOUT3	Analogue outputs	Analogue output 3	±10V full scale isolated output
5	COM1	Relay outputs	Relay 1 common	Maximum 1A / 24VDC
6	COM2	Relay outputs	Relay 2 common	Maximum 1A / 24VDC
7	COM3	Relay outputs	Relay 3 common	Maximum 1A / 24VDC
8	COM4	Relay outputs	Relay 4 common	Maximum 1A / 24VDC
9	AGND	Analogue input	Analogue input ground	Isolated ground
10	LIN2	Logic inputs	Logic input 2	Low < +3 V; High > +10.5 V; Max ±30 V
11	DGND	Logic inputs	Digital ground	For logic and speed pulse inputs, NOT isolated
12	SPD2	Speed pulse inputs	Speed input 2 (Low voltage)	Low < +2 V; High > +4 V; Max ±30 V
13	SPD1	Speed pulse inputs	Speed input 1 (High voltage)	Low < +5 V; High > +9 V; Max ±50 V
14	PIN	PI feedback controller	PI feedback controller input	Max ±20V isolated input
15	Agnd	Analogue outputs	Analogue output ground	Isolated ground
16	AOUT2	Analogue outputs	Analogue output 2	±10V full scale isolated output
17	NO1	Relay outputs	Relay 1 normally open contact	Maximum 1A / 24VDC
18	NO2	Relay outputs	Relay 2 normally open contact	Maximum 1A / 24VDC
19	NO3	Relay outputs	Relay 3 normally open contact	Maximum 1A / 24VDC
20	NO4	Relay outputs	Relay 4 normally open contact	Maximum 1A / 24VDC
21	-	-	-	Not connected
22	AIP	Analogue input	Analogue input	0–10 V full scale isolated input
23	LIN1	Logic inputs	Logic input 1	Low < +3 V; High > +10.5 V; Max ±30V
24	REG.GND	Power supply	Power supply ground (0 V)	
25	+24V	Power supply	+24 V power supply	
S	Shield			

PROTON STANDARD PARAMETER ACCESS PROTOCOL

This protocol provides access to individual parameters or blocks of parameters and is typically used in a production environment where the gauge is connected to a computer, PLC or similar device.

The parameters consist of 16-bit words (DW) and are divided into input and output groups:

- Input parameters may be read and written and are used to configure the gauge.
- Output parameters are read only and provide access to gauge status and measurement data. They may be requested to stream continuously from the gauge.

The Proton standard protocol RS-232 data format is:

Number of data bits	Parity	Number of stop bits	Flow control	Default baud rate
8	None	1	None	115200 bit / s

PARAMETER WORD FORMATS

Bit pattern words

Words defined as a bit pattern are formatted as 4 characters representing bits 15 – 0 in hexadecimal (0 – 9; A – F in capitals only), with bit 15 as the most significant bit. Unused or reserved bits are set to binary zero.

READING INPUT PARAMETERS

If a PC-based terminal program (such as MS Windows HyperTerminal) and keyboard is being used for RS-232 communication with the gauge, then the carriage return followed by line feed (<CR><LF>) sequence of commands can be replaced by a single press of the 'Enter ↵' key.

Reading a single input parameter

Use the following request format to read a single input parameter value:

Request format	?[Input parameter number]<CR><LF>
Response format	[Input parameter value]<CR><LF>

Reading a block of input parameters

Use the following request format to read a contiguous block of input parameter values:

Request format	?[Input parameter number][Space][Number of parameters]<CR><LF>
Response format	[Input parameter value]<CR><LF>

Only consecutive parameter values may be requested in a block parameter request.

WRITING INPUT PARAMETERS

Writing a single input parameter

Use the following request format to write a single input parameter value:

Request format	&[Input parameter number][Space][Input parameter value]<CR><LF>
Response format	[Input parameter value]<CR><LF>

The response is the post-write value of the input parameter; therefore a successful write is indicated by a response value that is identical to write request value.

READING OUTPUT PARAMETERS

If a PC-based terminal program (such as MS Windows HyperTerminal) and keyboard is being used for RS-232 communication with the gauge, then the carriage return followed by line feed (<CR><LF>) sequence of commands can be replaced by a single press of the 'Enter' key.

Reading a single output parameter

Use the following request format to read a single output parameter value:

Request format	~[Output parameter number][Space][Number of parameters]<CR><LF>
Response format	[Output parameter value]<CR><LF>

Reading a block of output parameters

Use the following request format to read a contiguous block of output parameter values:

Request format	~[Output parameter number][Space][Number of parameters] <CR><LF>
Response format	[Output parameter value]<CR><LF>

Only consecutive parameter values may be requested in a block parameter request.

Reading a continuous stream of a single output parameter

Use the following request format to read a single output parameter value as a continuous stream:

Request format	#[Output parameter number]<CR><LF>
Response format	[Output parameter value]<CR><LF>

The update rate for streaming output parameter values depends on both the data transfer rate and the data length. For a data transfer rate of 1200 bit / s and a data length of seven 8-bit characters per parameter value, approximately 10 individual parameter values can be transferred per second.

The data stream can be terminated by sending an escape character to the gauge (ASCII code decimal 27 or the 'Esc' key or the 'Ctrl'[' key combination on a PC keyboard).

Reading a continuous stream of a block of output parameters

Use the following request format to read a contiguous block of output parameter values as a continuous stream:

Request format	#[Output parameter number][Space][Number of parameters]<CR><LF>
Response format	[Output parameter value]<CR><LF>

Only consecutive parameter values may be requested in a block parameter request.

The update rate for streaming output parameter values depends on both the data transfer rate and the data length. For a data transfer rate of 1200 bit / s and a data length of seven 8-bit characters per parameter value, approximately 10 individual parameter values can be transferred per second.

The data stream can be terminated by sending an escape character to the gauge (ASCII code decimal 27 or the 'Esc' key or the 'Ctrl'[' key combination on a PC keyboard).

MODBUS PARAMETER ACCESS PROTOCOL

This protocol provides access to individual parameters or blocks of parameters and is typically used in a production environment where the Proton Products instrument is connected to a computer, Modbus connected PLC or similar device.

The parameters consist of 16-bit words (DW) and are divided into input and output groups:

- Input parameters may be read from and written to and are used to configure the instrument.
- Output parameters are read only and provide access to instrument status and measurement data.

1 word (DW) consists of 2 bytes
1 byte consists of 8 bits
4 bits are expressed by 1 hexadecimal digit (0-9, A-F)

Proton Products instruments use the following Modbus format:

Modbus format	RTU (Remote Terminal Unit)
Error check	CRC (Cyclic Redundancy Check)

The Modbus RTU frame format is as follows:

Modbus RTU frame format			
Name	Length	Function	
Start	> 3.5 characters	>3.5 characters of silence	
Address	1-byte	Slave (instrument) Modbus address	*
Function	1-byte	Modbus function code determines read or write operation	*
Data	Multiple bytes	Length and data dependent on function	*
CRC	2-bytes	Cyclic Redundancy Check for errors (not required for Modbus TCP) ⁺	
End	> 3.5 characters	>3.5 characters of silence	

⁺ Please refer to Modbus Protocol Standards documentation for the CRC error check field calculation. For Modbus protocol communications over TCP/IP networks via the Ethernet port (also known as **Modbus TCP**), the CRC error check field is not required as error checking is handled by the TCP/IP protocol.

Proton Products instruments support the following Modbus functions:

Modbus function			Instrument operation
Code		Name	
Dec	Hex		
03	03	Read Holding Registers	Read data from a block of consecutive input parameters.
06	06	Preset Single Register	Write data to a single input parameter.
16	10	Preset Multiple Registers	Write data to a block of consecutive input parameters.
04	04	Read Input Registers	Read data from a block of consecutive output parameters.

For Modbus protocol communications via the RS-232 serial port, the RS-232 data format is:

Number of data bits	Parity	Number of stop bits	Flow control	Default baud rate
8	None	1	None	115200 bit / s

INPUT PARAMETERS

DW	Bit	Comments	Unit	Range/Remark	Default value
DW0		System Mode			
	0	Measurement Mode		0=Solid;1=Glass;2=Helix;	0
	1				
	2				
	3	Unit		0=Metric;1=Imperial	0
	4	Shrinkage mode		0=Percentage;1=Abs	0
	5	SMFD tolerance mode		0=ABS;1=Percentage	0
	6	SMFD interval mode		0=Time;1=Length;	0
	7				
DW1					
DW2		Preset average diameter	1=0.01um/0.01mil	0~20000000	1000000
DW3					
DW4		Preset X diameter	1=0.01um/0.01mil	0~20000000	1000000
DW5					
DW6		Preset Y diameter	1=0.01um/0.01mil	0~20000000	1000000
DW7					
DW8		Preset Z diameter	1=0.01um/0.01mil	0~20000000	1000000
DW9					
DW10		Preset Ovality	1=0.01um/0.01mil	0~20000000	0
DW11					
DW12		Preset average upper limit	1=0.01um/0.01mil	0~20000000	50000
DW13					
DW14		Preset average lower limit	1=0.01um/0.01mil	0~20000000	50000
DW15					
DW16		Preset X upper limit	1=0.01um/0.01mil	0~20000000	50000
DW17					
DW18		Preset X lower limit	1=0.01um/0.01mil	0~20000000	50000
DW19					
DW20		Preset Y upper limit	1=0.01um/0.01mil	0~20000000	50000
DW21					
DW22		Preset Y lower limit	1=0.01um/0.01mil	0~20000000	50000
DW23					
DW24		Preset Z upper limit	1=0.01um/0.01mil	0~20000000	50000

DW	Bit	Comments	Unit	Range/Remark	Default value
DW25					
DW26		Preset Z lower limit	1=0.01um/0.01mil	0~20000000	50000
DW27					
DW28		Preset Ovality upper limit	1=0.01um/0.01mil	0~20000000	50000
DW29					
DW30		Preset Ovality lower limit	1=0.01um/0.01mil	0~20000000	50000
DW31					
DW32		Preset SMFD upper limit	1=0.01um/0.01mil/ 1=0.1%	0~20000000	50000
DW33					
DW34		Preset SMFD lower limit	1=0.01um/0.01mil/ 1=0.1%	0~20000000	50000
DW35					
DW36		Preset core diameter	1=0.01um/0.01mil	0~20000000	800000
DW37					
DW38		Diameter Average time	1=1ms	0~5000	1000
DW39		Shrinkage	1=1um{0.1mil}/ 1=0.1%	0~10000	0
DW40		Pitch for helix	1=10um{1=1mil}	1~65535	1000
DW41		Core number for helix	1=1	1~100	2
DW42		Flaw interval time	1=0.001s/1=0.001m{1 =0.001ft}	1~65535	100
DW43		SMFD reference diameter average time	1=1ms	1~1000	100
DW44		SMFD measurement average time	1=0.1ms	1~100	10
DW45		Start speed for SMFD detection	1=1m/min{1=1ft/min}	1~65535	0
DW46		Start Length for SMFD	1=1m{ft}		0
DW47		Start Length for Diameter over tolerance	1=1m{ft}		0
DW48		Reset length / running maximum and minimum /flaw counting		1=Reset;other=Invalid	0
DW49		Logic input			
	0	Logic input 1 function		0=Reset;1=External Alarm 1;2=External Alarm 2;3=Line rev direction	0
	1				
	2	Logic input 2 function		0=Reset;1=External Alarm 1;2=External Alarm 2;3=Line rev direction	1
	3				
	4	The polarity of logic input 1		0=Active low; 1=Active high	0
	5	The polarity of logic input 2		0=Active low; 1=Active high	0
DW50		Relays			
	0	Relay 1 function		0=Gauge OK;1=Over any limit; 2=Under any limit;3=Flaw;4=Lump;5=N eck, 6=Over Average Upper limit;7=Under	0
	1				
	2				

DW	Bit	Comments	Unit	Range/Remark	Default value
	3	Relay 2 function		Average Lower limit;8=Over X Upper limit;9=Under X Lower limit;10=Over Y Upper limit;11=Under Y Lower limit;12=Over Z Upper limit;13=Under Z Lower limit;14=Over ovality Upper limit;15=Window Dirty	1
	4				
	5				
	6				
	7				
	8	Relay 3 function			2
	9				
	10				
	11				
	12	Relay 4 function			3
	13				
	14				
	15				
DW51		Relay closure time	1=1ms	1~5000	100
DW52		Source of line speed		0=Preset;1=Pulse;2=Analogue	0
DW53		Preset line speed	1=1m/min{1=ft/min}	0~65535	1000
DW54		Full scale/gain of line speed	1=1m/min{1=1ft/min)		1000
DW55			or 1=0.0001pulse/meter{ 1=0.0001pulse/ft)		
DW56	0~7	Control switch		0=HOLD;1=ON;2=RESET	0
	8	Control output polarity		0=Normal;1=Reverse	0
DW57		PID Start Speed	1=1m/min{1=1ft/min}		50
DW58		PID output Range	1=1%	0~50%	50
DW59		Extruder response time	1=1s	0~999s	1
DW60		Distance from gauge head to extruder	1=0.1m{1=0.1ft}	1~10000	10
DW61		I gain	1=1%	0~100	50
DW62		P gain	1=1%	0~100	50
DW63		Analogue O/P			
	0	Analogue O/P 1		0=Average ;1=X;2=Y;3=Z ;4=Ovality;5=Average error;6=X error;7=Y error;8=Z error;9=Ovality error;	0
	1				
	2				
	3				
	4	Analogue O/P 2			1
	5				
	6				
	7				
	8	Analogue O/P 3			2

DW	Bit	Comments	Unit	Range/Remark	Default value
	9				
	10				
	11				
	12				
	13				
	14				
	15	Response speed		0=Instant;1=Averaged	0
DW64		Full scale for analogue O/P 1	1=0.01um/0.01mil	0~20000000	1000000
DW65					
DW66		Full scale for analogue O/P 2	1=0.01um/0.01mil	0~20000000	1000000
DW67					
DW68		Full scale for analogue O/P 3	1=0.01um/0.01mil	0~20000000	1000000
DW69					
DW70		SPC switch		0=OFF;1=ON	0
DW71		Statistics time	1=1s{1=1m{ft}}	1~65535	10
DW72		Statistics Mode		0=Time;1=Length;2=One Reel	0
DW73		FFT sampling rate		0=OFF;1=1Hz;2=3Hz;3=10Hz;4=30Hz;5=100Hz;6=300Hz;7=1KHz	0
DW74		Hold time for external alarm	1=1ms	1~5000	300
DW75					
DW76					
DW77					
DW78					
DW79					
DW80		Profibus ID	1=1	0~125	4
DW81		DeviceNet address		0~63	3
DW82		DeviceNet baud rate		0=125K;1=250K;2=500K	0
DW83		CAN BUS address	1=1	0~255	17
DW84		CAN BUS baud rate		0=250;1=500;2=1000;other=500	0
DW85		RS232 baud rate		0=4800;1=9600;2=19200;3=38400;4=115200	4
DW86		RS232 protocol		0=Modbus;1=Proton;2=SLP;3=Print	0
DW87		RS422/RS485 protocol		0=Modbus;1=Proton;2=SLP	0
DW88		RS422/RS485 baud rate		0=4800;1=9600;2=19200;3=38400;4=115200;5=250K;6=500K;7=1M	4

DW	Bit	Comments	Unit	Range/Remark	Default value
DW89		Modbus ID	1=1	0~255	1
DW90		ETH DHCP		0=Disable;1=Enable	0
DW91		iBus DHCP		0=Disabled;1=Enabled	0
DW92		ETH IP address	xx.xx.xx.xx		C0A801AA (192.168.1.170)
DW93					
DW94		Rev			
DW95					
DW96		Subnet mask	xx.xx.xx.xx		255.255.255.0
DW97					
DW98		Gateway	xx.xx.xx.xx		C0A80101(192. 168.1.1)
DW99					
DW100		iBus IP Address			C0A80165(192. 168.1.171)
DW101					
DW102		iBus Subnet Mask			FFFFFF00(255. 255.255.0)
DW103					
DW104		iBus Gateway			C0A80101(192. 168.1.1)
DW105					
DW106		Destination IP address for UDP (last section only)		Port number fixed to 1111	2
DW107		Interval time of UDP data output	1=1ms	0~5000(0=disable UDP output)	0
DW108		Little Endian/Big Endian		1=Little Endian;0=Big Endian	1
DW109		CAN terminator	1=1	0=OFF;1=ON	1
DW110		CAN BUS Protocol		0=Ver2.0;1=old	
DW111					
DW112					
DW113					
DW114					
DW115					
DW116		Diameter compensation coefficient	1=0.0001		10000
DW117		WIFI switch			
DW118		Gain of analogue O/P1	1=1		10000
DW119		Zero of analogue O/P1	1=1		0
DW120		Gain of analogue O/P2	1=1		10000
DW121		Zero of analogue O/P2	1=1		0
DW122		Gain of analogue O/P3	1=1		10000
DW123		Zero of analogue O/P3	1=1		0
DW124					

DW	Bit	Comments	Unit	Range/Remark	Default value
DW125		Gain of analogue input	1=1		10000
DW126		Zero of analogue input	1=1		0
DW127		Parameter Group Number		1~50; 0 is not used	1
DW128		Parameter Group Control		1=Load parameters from specified group;2=Save current parameters to specified group	0
DW129					

OUTPUT PARAMETERS

DW	Bit	Comments	Units	Range/Remark
DW0	0	Measurement Mode		0=Solid;1=Glass;2=Helix
	1			
	2			
	3	Unit		0=Metric;1=Imperial
	4	Shrinkage mode		0=Percentage;1=ABS
	5	High Resoulation for Diameter		0=OFF;1=ON
	6	Over average upper limit		0=OK;1=Over limit
	7	Under average lower limit		0=OK;1=Over limit
	8	Over X upper limit		0=OK;1=Over limit
	9	Under X lower limit		0=OK;1=Over limit
	10	Over Y upper limit		0=OK;1=Over limit
	11	Under Y lower limit		0=OK;1=Over limit
	12	Over Z upper limit		0=OK;1=Over limit
	13	Under Z lower limit		0=OK;1=Over limit
	14	Over ovality upper limit		0=OK;1=Over limit
	15	Under ovality lower limit		0=OK;1=Over limit
DW1	0	Res		
	1	No reading		0=OK;1=No Reading
	2	No object		0=OK;1=No Object
	3	Window dirty		0=OK;1=Window Dirty
	4	Line speed too low in helix mode		0=OK;1=too low
	5	Line speed too high in helix mode		0=OK;1=too high
	6	Gauge too hot		0=OK;1=Gauge Hot
	7			
	8	External Alarm 1		0=OK;1=Alarm
	9	External Alarm 2		0=OK;1=Alarm

DW	Bit	Comments	Units	Range/Remark
DW2		Average diameter/Envelop	1=0.01um{mil}	0~20000000
DW3				
DW4		X diameter	1=0.01um{mil}	0~20000000
DW5				
DW6		Y diameter	1=0.01um{mil}	0~20000000
DW7				
DW8		Z diameter	1=0.01um{mil}	0~20000000
DW9				
DW10		Ovality	1=0.01um{mil}	0~20000000
DW11				
DW12		Average error/Envelop Error	1=0.01um{mil}	-20000000~20000000
DW13				
DW14		X error	1=0.01um{mil}	-20000000~20000000
DW15				
DW16		Y error	1=0.01um{mil}	-20000000~20000000
DW17				
DW18		Z error	1=0.01um{mil}	-20000000~20000000
DW19				
DW20		Ovality error	1=0.01um{mil}	-20000000~20000000
DW21				
DW22		Latest lump value	1=0.01um{mil}	-20000000~20000000
DW23				
DW24		Latest lump positon	1=0.001m{1=0.001ft}	0~100000000
DW25				
DW26		Latest neck value	1=0.01um{mil}	-20000000~20000000
DW27				
DW28		Latest neck position	1=0.001m{1=0.001ft}	0~100000000
DW29				
DW30		Lump count	1=1	0~65535
DW31		Neck count	1=1	0~65535
DW32		Running maximum diameter	1=0.01um{mil}	0~20000000
DW33				
DW34		Running minimum diameter	1=0.01um{mil}	0~20000000
DW35				
DW36		Running average diameter	1=0.01um{mil}	0~20000000
DW37				
DW38		Cable position on X axis	1=1%	0~±100
DW39		Cable position on Y axis	1=1%	0~±100
DW40		Cable position on Z axis	1=1%	0~±100
DW41		Line speed	1=0.1m/min{1=1ft/min}	-32768~32767

DW	Bit	Comments	Units	Range/Remark
DW42		Length	1=0.001m{1=0.001ft}	
DW43				
DW44		Statistics		
	0	Normal distribution		0=No;1=Yes
	1	Not used		
	2	SPC is available		0=No;1=Yes
DW45		Statistics remain time	1=1s{1=1m{ft}}	0~65535
DW46		Standard deviation	1=0.01um{mil}	0~20000000
DW47				
DW48		Maximum diameter	1=0.01um{mil}	0~20000000
DW49				
DW50		Minimum diameter	1=0.01um{mil}	0~20000000
DW51				
DW52		Mean diameter	1=0.01um{mil}	0~20000000
DW53				
DW54		Chi of normal distribution	1=1%	Not used
DW55		Cp	1=1%	
DW56		Cpk	1=1%	
DW57		FFT remain time	1=1s	
DW58		Control status		0=OFF;1=ON;2=Reset;3=Ready
DW59		Control output	1=0.01%	0~±5000
DW60				
DW61				
DW62		Communication Bus Type		0=NONE;1=PROFIBUS;2=Ethernet /IP;3=PROFINET
DW63		Res		
DW64		IP address for ETH	xx.xx.xx.xx	
DW65				
DW66		IP address for iBUS	xx.xx.xx.xx	
DW67				
DW68		Sub net Mask	xx.xx.xx.xx	
DW69				
DW70		Gateway	xx.xx.xx.xx	
DW71				
DW72		Gauge temperature	1=0.1℃	

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